A PROJECT REPORT

ON

MACHINE LEARNING-BASED PREDICTION OF CHILDHOOD VACCINATION IN INDIA: EVIDENCE FROM NFHS-5 (2022)

SUBMITTED IN THE PARTIAL FULFILMENT OF THE REQUIREMENTS FOR THE AWARD OF THE DEGREE OF

MASTER OF SCIENCE

IN

DATA SCIENCE

SUBMITTED BY:

ABDUL BASITH A P	**25DSMSA101**
SHADHA O	**25DSMSA102**
MOHAMMAD ASIM	**25DSMSA104**
MOHD MUKEEM	**25DSMSA105**

UNDER THE SUPERVISION OF

DR. IRFAN ALI
DR. FAIZ NOOR KHAN YUSUFI
DR. SHAZIA FARHIN

DEPARTMENT OF STATISTICS & OPERATIONS RESEARCH

ALIGARH MUSLIM UNIVERSITY

ALIGARH (202002) U.P., INDIA

2025-26

DEPARTMENT OF STATISTICS &
OPERATIONS RESEARCH,
ALIGARH MUSLIM UNIVERSITY,
ALIGARH-202002 (U.P.) INDIA

Certificate

This is to certify that the dissertation entitled **"Machine Learning-Based Prediction of Childhood Vaccination in India: Evidence from NFHS-5 (2022)"** has been carried out by the following group of students of **Master of Science (Data Science),** First Year, 2025–26, under our joint supervision for the course "**DSMSMJ8D01-Dissertation**".

SUBMITTED BY:

ABDUL BASITH A P	**25DSMSA101**
SHADHA O	**25DSMSA102**
MOHAMMAD ASIM	**25DSMSA104**
MOHD MUKEEM	**25DSMSA105**

Dr. Irfan Ali
(Supervisor)

Dr. Faiz Noor Khan Yusufi
(Co-Supervisor)

Dr. Shazia Farhin
(Co-Supervisor)

ACKNOWLEDGEMENT

All praise and thanks are to The Almighty, the Most Merciful and Omniscient, whose blessings enabled us to come across the completion of this work in its present form.

We are thankful to the Chairperson of our department*, **Prof. Bushra Husain***, for providing us with all the necessary facilities that were imperative for the successful completion of this project work.

It is our privilege to express our sincere thanks and deep sense of gratitude to our supervisor ***Dr. Irfan Ali*** and co-supervisors ***Dr. Faiz Noor Khan Yusufi*** and ***Dr. Shazia Farhin*** Department of Statistics and Operations Research, AMU, Aligarh for their guidance, efforts, valuable insight and support throughout the project. Their expert advice and encouragement have been invaluable to our success.

We would also like to express our special gratitude to our parents and well- wishers for their prayers and for providing us with moral support to pursue our studies.

ABDUL BASITH A P

SHADHA O

MOHAMMAD ASIM

MOHD MUKEEM

DECLARATION

We hereby declare that the project entitled **"Machine Learning-Based Prediction of Childhood Vaccination in India: Evidence from NFHS-5 (2022)"** is an original academic work carried out by us under the supervision of ***Dr. Irfan Ali*** and along with our co-supervisors ***Dr. Faiz Noor Khan Yusufi***, and ***Dr. Shazia Farhin,*** Department of Statistics and Operations Research, AMU, Aligarh.

The study utilizes secondary data obtained from the ***National Family Health Survey-5 (NFHS-5)*** dataset, conducted in India and made available through the ***Demographic and Health Survey (DHS)*** program. The dataset has been used strictly for academic and research purposes. The findings, interpretations, and conclusions presented in this project are intended solely for academic and research purposes and should not be considered as clinical recommendations, healthcare policies, or medical decision-making tools.

We further declare that this project has not been submitted previously, either in part or in full, to any university or institution for the award of any degree, diploma, or certification. All sources of data, literature, references, and software tools used in this study have been properly acknowledged.

ABDUL BASITH A P

SHADHA O

MOHAMMAD ASIM

MOHD MUKEEM

ABSTRACT

Childhood vaccination is widely recognized as one of the most effective and cost-efficient public health interventions for reducing child morbidity and mortality associated with vaccine-preventable diseases. Immunization plays a crucial role in protecting children from infectious diseases and contributes to improved community health through herd immunity and reduced disease transmission. Adequate vaccination coverage significantly enhances child survival and supports overall public health and socioeconomic development. In developing countries such as India, ensuring complete childhood immunization remains a major public health priority because of the large population size and persistent socioeconomic and regional disparities. To address these challenges, the Government of India introduced the Universal Immunization Programme (UIP), one of the world's largest public health initiatives, targeting approximately 26 million newborns annually. Despite considerable progress in immunization coverage, inequalities in access to healthcare services and socioeconomic conditions continue to affect vaccination uptake. According to the National Family Health Survey-5 (NFHS-5, 2019–21), approximately 63% of children aged 12–23 months in India were fully vaccinated, leaving a considerable proportion of children vulnerable to vaccine-preventable diseases and associated health complications.

The present study aimed to predict childhood vaccination status among children aged 12–23 months in India using machine learning techniques and identify the major determinants associated with vaccination uptake. Additionally, the study sought to examine patterns and disparities in vaccination coverage among different demographic and socioeconomic groups. Data for this study were obtained from the National Family Health Survey-5 (NFHS-5) Children's Recode dataset (IAKR7EFL), a nationally representative survey containing detailed information on child health, maternal characteristics, healthcare utilization, and household

socioeconomic conditions. After applying inclusion and exclusion criteria and conducting data preprocessing procedures, a final sample of 12,377 children aged 12–23 months was included in the analysis.

A comprehensive Exploratory Data Analysis (EDA) was conducted to understand the characteristics and distribution of variables within the dataset. Descriptive and visualization techniques were used to identify patterns, trends, and relationships among variables. Data preprocessing steps included handling missing values, feature selection, encoding categorical variables, and data transformation where necessary to improve data quality and analytical efficiency. These procedures ensured that the dataset was suitable for machine learning model development and predictive analysis.

Five supervised machine learning algorithms were implemented and compared in this study: Logistic Regression, K-Nearest Neighbors (KNN), Support Vector Machine (SVM), Random Forest, and XGBoost. These algorithms were selected because of their effectiveness and broad applicability in classification problems within healthcare and public health research. Model performance was evaluated using multiple performance metrics, including accuracy, sensitivity, specificity, precision, F1-score, Area Under the Receiver Operating Characteristic Curve (AUC–ROC), and Cohen's Kappa coefficient, allowing a comprehensive assessment of predictive performance.

Among all machine learning algorithms, XGBoost demonstrated the best predictive performance. The model achieved an accuracy of 86.38%, an AUC value of 94.27%, an F1-score of 89.54%, and a Cohen's Kappa coefficient of 70.07%, indicating strong predictive capability and substantial agreement. Feature importance analysis identified child age in months, health worker visits, household wealth index, birth order, and antenatal care (ANC) visits as the most influential predictors of childhood vaccination status. The exploratory findings further revealed considerable disparities in vaccination coverage across population groups. Rural children exhibited lower vaccination rates compared with urban children. Similarly, children from poorer households demonstrated substantially lower vaccination coverage compared with those from wealthier households. Maternal education was also found to be an important determinant, as children born to mothers with lower educational attainment showed reduced vaccination uptake. These findings indicate the continued influence of socioeconomic inequalities and healthcare accessibility on childhood immunization outcomes.

Overall, the findings of this study demonstrate the potential of machine learning techniques in predicting childhood vaccination status and identifying key determinants of immunization uptake. The results provide valuable evidence for policymakers and public health practitioners to design targeted interventions and strengthen existing immunization programs, including Mission Indradhanush with the goal of improving vaccination coverage and reducing inequalities in child healthcare outcomes across India.

TABLE OF CONTENTS

LIST OF TABLES

LIST OF FIGURES

CHAPTER 1:

INTRODUCTION

1.1 Background

Immunization Vaccination is universally recognized as one of the most successful and cost-effective public health interventions in human history. According to the World Health Organization, immunization prevents approximately 2–3 million deaths annually worldwide, and with improved global coverage, an additional 1.5–2 million deaths could be prevented each year. Vaccines provide protection against several life-threatening infectious diseases such as tuberculosis, diphtheria, pertussis, tetanus, poliomyelitis, measles, and hepatitis B, particularly among children under five years of age. Consequently, immunization has become a cornerstone of global child survival and disease prevention strategies.

In India, childhood immunization has been a major public health priority since the launch of the Universal Immunization Programme (UIP) in 1985. The UIP is one of the world's largest public health initiatives and currently targets millions of newborns and pregnant women annually against vaccine-preventable diseases. Over the years, the Government of India has strengthened immunization efforts through various supplementary initiatives such as Mission Indradhanush (2014) and Intensified Mission Indradhanush (2017), which were specifically designed to increase vaccine coverage among underserved and hard-to-reach populations. These programmes focus particularly on regions characterized by low healthcare accessibility, socioeconomic deprivation, and poor health awareness.

Despite substantial improvements in vaccination coverage over the past decade, inequalities in immunization outcomes continue to persist across Indian states and demographic groups. Evidence from the National Family Health Survey-5 (NFHS-5, 2019–21), one of the largest nationally representative health surveys in India, reveals considerable disparities in full vaccination coverage among children aged 12–23 months. Vaccination rates vary significantly according to wealth status, maternal education, rural–urban residence, caste, religion, and regional differences. Children belonging to economically disadvantaged households, rural communities, and less educated families are more likely to remain partially vaccinated or completely unvaccinated. Such disparities create clusters of vulnerable populations that are susceptible to outbreaks of vaccine-preventable diseases.

Traditionally, statistical methods such as logistic regression have been widely employed to study determinants of childhood immunization. While these methods are useful for identifying linear relationships and statistically significant predictors, they often face limitations when handling high-dimensional datasets, complex nonlinear interactions, and large-scale survey data. Furthermore, traditional approaches may not always provide sufficiently accurate predictive capabilities for identifying children at risk of incomplete vaccination at the individual level.

In recent years, Machine Learning (ML) techniques have emerged as powerful analytical tools in healthcare research and predictive public health analytics. ML algorithms possess the ability to learn complex patterns from large datasets, handle nonlinear relationships among variables, and generate highly accurate predictive models. In the context of immunization research, ML approaches can help identify hidden patterns, classify vulnerable groups, and assist policymakers in designing targeted interventions for improving vaccine coverage.

Several international studies have demonstrated the effectiveness of ML techniques in predicting childhood vaccination status and other healthcare outcomes. However, despite the increasing availability of nationally representative health datasets such as NFHS-5, relatively few studies in India have conducted a comprehensive comparative evaluation of multiple ML algorithms for predicting childhood vaccination coverage. Most existing Indian studies rely predominantly on conventional statistical analyses, leaving significant scope for advanced predictive modelling approaches.

This dissertation seeks to address this research gap by applying and comparing multiple Machine Learning algorithms—including Logistic Regression, K-Nearest Neighbors (KNN), Support Vector Machine (SVM), Random Forest, and XGBoost, —to predict childhood vaccination status using the NFHS-5 children's recode dataset. In addition, the study incorporates extensive Exploratory Data Analysis (EDA) to understand the distribution, trends, and relationships among socioeconomic, demographic, and maternal health variables associated with childhood vaccination.

1.2 Statement of the Problem

Although India has made notable progress in expanding immunization coverage through various national health programmes, a significant proportion of children aged 12–23 months remain under-vaccinated or completely unvaccinated. According to the NFHS-5 dataset used in this study, approximately 37 percent of children are not fully vaccinated, representing millions of vulnerable children across the country. Incomplete immunization not only increases the risk of childhood morbidity and mortality but also weakens herd immunity and places additional pressure on the public healthcare system.

One of the major challenges faced by health authorities and policymakers is the inability to accurately identify children who are at higher risk of missing vaccinations. Traditional analytical approaches primarily focus on association analysis and often lack strong predictive capability at the individual level. As a result, public health interventions may not always reach the most vulnerable populations efficiently.

Moreover, vaccination outcomes are influenced by multiple interconnected factors such as maternal education, household wealth, healthcare accessibility, media exposure, geographic location, and sociocultural characteristics. These variables frequently interact in complex and nonlinear ways that conventional statistical techniques may fail to capture effectively.

Machine Learning methods offer a promising alternative due to their superior capability in handling large datasets, identifying hidden patterns, and improving predictive accuracy. International evidence from countries such as Ethiopia, Bangladesh, Pakistan, and the United States suggests that ML-based predictive models can significantly enhance healthcare targeting and decision-making. However, there remains a lack of comprehensive comparative studies applying multiple ML algorithms to India's NFHS-5 vaccination data.

Therefore, this study aims to fill this gap by developing and comparing several ML classification models for predicting childhood vaccination status in India. The study further seeks to identify the most influential predictors of vaccination and provide evidence-based recommendations for improving immunization coverage among vulnerable populations.

1.3 Objectives of the Study

1.3.1 General Objective

The primary objective of this study is to apply and compare various Machine Learning algorithms to predict childhood vaccination status among children aged 12–23 months using the NFHS-5 children's recode dataset and to identify the most significant determinants influencing vaccination coverage in India.

1.3.2 Specific Objectives

The specific objectives of the study are as follows:

- To conduct a comprehensive Exploratory Data Analysis (EDA) of the NFHS-5 dataset to examine the distribution and relationships of variables associated with childhood vaccination.
- To preprocess the dataset by handling missing values, encoding categorical variables, standardizing features where necessary, and addressing class imbalance using SMOTE (Synthetic Minority Over-sampling Technique) when required.
- To develop, train, and evaluate multiple Machine Learning classification models—including Logistic Regression, K-Nearest Neighbors (KNN), Support Vector Machine (SVM), Random Forest, and XGBoost—for predicting full vaccination status among children.
- To compare the predictive performance of these models using evaluation metrics such as Accuracy, Precision, Recall (Sensitivity), Specificity, F1-score, and Area Under the ROC Curve (AUC-ROC).
- To identify the most influential socioeconomic, demographic, maternal, and household-level factors associated with childhood vaccination coverage.
- To provide evidence-based policy recommendations for improving immunization coverage and reducing disparities in vaccination outcomes across India.

1.4 Research Questions

The study seeks to answer the following research questions:

1. Which Machine Learning algorithm performs best in predicting full vaccination status among children aged 12–23 months using NFHS-5 data?
2. What are the most significant socioeconomic, demographic, maternal, and household factors influencing childhood vaccination in India?

3. How do disparities in vaccination coverage vary across wealth groups, maternal education levels, place of residence, and geographic regions?
4. Can Machine Learning models provide improved predictive capability compared to traditional statistical approaches in identifying at-risk children?
5. What policy interventions can be recommended based on the findings from Exploratory Data Analysis and Machine Learning models?

1.5 Scope and Significance

This study utilizes secondary data obtained from the NFHS-5 children's recode dataset, comprising children aged 12–23 months from all states and union territories of India. The analysis focuses specifically on childhood vaccination status and its associated determinants. The study combines Exploratory Data Analysis with advanced Machine Learning techniques to generate predictive insights relevant to public health policy and healthcare planning.

The significance of this study lies in both its academic contribution and practical policy relevance. From an academic perspective, the study contributes to the growing interdisciplinary field of data science and public health by demonstrating the application of Machine Learning methods in healthcare prediction using large-scale national survey data. It also enriches existing literature on childhood immunization determinants in India.

From a policy perspective, the study provides valuable insights for government agencies, healthcare administrators, and public health organizations. By identifying high-risk populations and key determinants of incomplete vaccination, the findings can assist programmes such as Mission Indradhanush, the National Health Mission (NHM), and frontline healthcare systems in designing targeted interventions. The predictive models developed in this study may potentially serve as decision-support tools for identifying vulnerable children and optimizing resource allocation in immunization campaigns.

Additionally, the study highlights the potential role of Artificial Intelligence and Machine Learning in strengthening healthcare delivery systems in developing countries such as India.

CHAPTER 2:

LITERATURE REVIEW

2.1 Childhood Vaccination — Global and Indian Context

The Childhood vaccination is widely regarded as one of the greatest achievements in the history of modern medicine and public health. Vaccines have played a transformative role in reducing childhood morbidity and mortality by preventing the spread of infectious diseases that once caused millions of deaths worldwide. The successful eradication of smallpox in 1980 through a coordinated global immunization campaign remains one of the most significant milestones in public health history. Similarly, global vaccination initiatives have led to dramatic reductions in diseases such as poliomyelitis, measles, diphtheria, pertussis, tetanus, and hepatitis B. According to the World Health Organization, immunization currently saves approximately 2–3 million lives annually, and this number could increase substantially with improved vaccine coverage and accessibility.

Vaccination not only protects individuals from infectious diseases but also contributes to the development of herd immunity, thereby reducing the transmission of diseases within communities. High vaccination coverage is particularly important for protecting vulnerable populations such as infants, immunocompromised individuals, and economically disadvantaged groups who may have limited access to healthcare services. Consequently, childhood immunization programmes are considered essential components of national healthcare systems and sustainable public health development.

Globally, international organizations such as the WHO, UNICEF, and Gavi, the Vaccine Alliance, have invested heavily in expanding vaccine accessibility in low- and middle-income countries. Despite substantial progress, significant disparities in vaccination coverage continue to exist across countries and regions. Factors such as poverty, inadequate healthcare infrastructure, geographical barriers, conflict situations, misinformation, and lack of awareness often hinder the achievement of universal immunization coverage. The COVID-19 pandemic further disrupted routine immunization services in many countries, increasing the risk of outbreaks of vaccine-preventable diseases.

In the Indian context, childhood vaccination has been a central public health priority for several decades. India launched the Expanded Programme on Immunization (EPI) in 1978, which was later expanded into the Universal

Immunization Programme (UIP) in 1985. The UIP is currently one of the largest immunization programmes in the world and aims to provide free vaccination services against major vaccine-preventable diseases to children and pregnant women across the country. The programme includes vaccines against tuberculosis, polio, diphtheria, pertussis, tetanus, hepatitis B, measles, rubella, and several other infectious diseases.

To further strengthen immunization coverage, the Government of India introduced targeted initiatives such as Mission Indradhanush in 2014 and Intensified Mission Indradhanush in 2017. These programmes specifically focus on reaching partially vaccinated and unvaccinated children living in underserved, remote, and socioeconomically disadvantaged areas. Through these initiatives, the government has attempted to reduce inequalities in immunization access and improve overall child health outcomes.

For a child aged 12–23 months to be classified as “fully vaccinated” in the Indian context, the child must receive all recommended basic vaccines within the prescribed schedule. According to NFHS-5 criteria, this includes one dose of Bacillus Calmette–Guérin (BCG) vaccine at birth; three doses of Pentavalent vaccine protecting against Diphtheria, Pertussis, Tetanus, Hepatitis B, and Haemophilus influenzae type b (Hib); three doses of Oral Polio Vaccine (OPV); and one dose of Measles or Measles–Rubella (MR) vaccine. The NFHS-5 survey records vaccination information using vaccination cards, health facility records, and maternal recall in cases where documentary evidence is unavailable.

Despite continuous improvements in healthcare delivery and immunization campaigns, India still faces considerable challenges in achieving universal vaccination coverage. The National Family Health Survey-5 (2019–21) reported a national full vaccination coverage rate of approximately 76.4 percent among children aged 12–23 months. However, substantial inequalities remain across states, socioeconomic groups, and demographic categories. Certain states such as Goa, Sikkim, and Puducherry report very high vaccination coverage, whereas states such as Nagaland, Meghalaya, and Arunachal Pradesh continue to exhibit relatively low coverage levels.

These disparities are influenced by multiple factors including maternal education, household income, access to healthcare facilities, place of residence, birth order, media exposure, cultural beliefs, and regional healthcare infrastructure. Children from rural areas, poorer households, and socially marginalized communities are more likely to experience incomplete immunization. Furthermore, issues such as vaccine hesitancy, misinformation, transportation barriers, and shortages of healthcare personnel continue to affect vaccine uptake in several regions.

In recent years, increasing attention has been directed toward the application of data-driven approaches and Machine Learning techniques in public health research. Large-scale datasets such as NFHS-5 provide an opportunity to identify hidden patterns and determinants associated with vaccination behaviour. By utilizing advanced analytical techniques, researchers and policymakers can better understand high-risk populations, predict vaccination outcomes, and design targeted interventions to improve immunization coverage.

Therefore, understanding the global and Indian context of childhood vaccination is essential for evaluating the determinants of vaccine uptake and developing effective predictive models for identifying children at risk of incomplete immunization. This study contributes to that objective by integrating public health knowledge with Machine Learning methodologies using nationally representative NFHS-5 data.

2.2 Determinants of Childhood Vaccination Coverage

2.2.1 Maternal Factors

Mother's education is consistently the strongest individual-level predictor of childhood vaccination in India. Educated mothers are more likely to be aware of the vaccination schedule, understand the benefits of immunization, and proactively seek health services for their children (Singh & Sahu, 2018; Shrivastwa et al., 2015). The NFHS-5 data in this study shows a clear gradient: children born to mothers with higher education have substantially higher vaccination rates than those born to uneducated mothers.

Antenatal Care (ANC) visits are another powerful predictor. Mothers who attend 4 or more ANC visits receive counselling on immunization, are more likely to deliver at a health facility, and develop stronger links with the healthcare system that facilitate follow-up immunizations (Tamirat & Sisay, 2019). The study dataset shows that 52.5% of children have mothers who had adequate ANC visits (4 or more).

Mother's age also plays a role — older mothers with prior experience of childbearing tend to have better health-seeking behaviour. The proportion of mothers aged above 35 years in the dataset is 16.1%.

2.2.2 Healthcare Access Factors

Place of delivery is one of the most powerful predictors of childhood vaccination. Children born at government or private health facilities are immediately enrolled in the immunization system and receive birth-dose vaccines. The study data shows that 82.4% of children in the sample were born at health facilities (institutional delivery), which is a significant improvement from previous NFHS rounds. However, the remaining 17.6% born at home face a substantially elevated risk of missing all vaccines.

Health worker visits — where ASHA (Accredited Social Health Activist) workers or ANMs (Auxiliary Nurse Midwives) visit households — are critical for reaching children who would otherwise be missed. In the study dataset, 56.3% of children received a health worker visit.

2.2.3 Socioeconomic Factors

Household wealth index (coded 1=poorest to 5=richest in NFHS) is strongly associated with vaccination coverage. Wealthier families can better afford transportation to health centers, are more likely to be educated, and have better access to information. The study EDA shows vaccination rates rising progressively from 45.7% in the poorest quintile to over 76.2% in the richest quintile. This wealth-vaccination gradient represents a major equity concern for India's public health system.

Place of residence (urban vs. rural) also significantly impacts vaccination. Urban children benefit from greater proximity to health facilities, better infrastructure, and higher awareness. The study data shows urban children have a vaccination rate of 72.3% compared to 59.8% for rural children. Given that 74.3% of the study sample is rural, this gap represents a massive public health challenge.

Religion, caste, and household head's sex also matter. Children from marginalized caste groups and those from female-headed households may face different patterns of health service access (Kapoor & Singh, 2014).

2.2.4 Child-Level Factors

Birth order has a clear negative association with vaccination — higher birth-order children (4th, 5th, and above) tend to have lower vaccination rates, possibly because parents allocate less attention and resources to later children. Birth interval and child's sex also play roles, though sex-based disparities in vaccination have narrowed over successive NFHS rounds.

2.3 Machine Learning in Public Health

Machine learning is a subfield of artificial intelligence where computer algorithms learn patterns from data to make predictions or decisions without being explicitly programmed. In public health, ML has been applied to predict disease incidence, patient outcomes, treatment adherence, child mortality, malnutrition, and health service utilization. ML is particularly well-suited to large health survey datasets like NFHS-5, where traditional methods may struggle with high dimensionality, non-linearity, and complex variable interactions.

Supervised learning — where algorithms are trained on labelled data with a known outcome — is most relevant for this study. The outcome (fully vaccinated vs. not fully vaccinated) is known for each record, making this a binary classification problem. Multiple supervised classifiers can be compared and the best-performing algorithm selected for deployment.

2.4 ML Algorithms Used in This Study

- **Binary Classification**

Binary classification is used when the response variable consists of only two categories or classes. The objective of binary classification is to predict whether an observation belongs to one of two possible outcomes based on predictor variables. Examples include outcomes such as *Yes/No*, *Vaccinated/Not Vaccinated*, or *Disease Present/Disease Absent*. Binary classification models estimate the probability of occurrence of one class and classify observations accordingly. In this study, binary classification techniques were employed when the response variable had two outcome categories.

- **Multi-class (Multinomial) Classification**

  Multi-class or multinomial classification is used when the response variable contains more than two categories. The objective is to classify observations into one among several possible outcome classes. Unlike binary classification, where only two classes are considered, multinomial classification simultaneously handles multiple categories. Examples include outcomes such as *No Vaccination*, *Partial Vaccination*, and *Full Vaccination*. These techniques estimate probabilities for multiple classes and assign observations to the class with the highest predicted probability.

  In this study, multinomial classification techniques were applied when the outcome variable consisted of more than two categories.

2.4.1 Logistic Regression

Logistic Regression is a supervised machine learning algorithm used for binary classification problems where the outcome variable has two possible categories, such as vaccinated/non-vaccinated, yes/no, or disease/no disease. The model estimates the probability that a particular observation belongs to a specific class based on a set of predictor variables.

Unlike linear regression, logistic regression uses the logistic sigmoid function to transform predicted values into probabilities ranging between 0 and 1. The relationship between the dependent variable and independent variables is represented through the logit function. The coefficients obtained from the model indicate the direction and strength of association between predictor variables and the outcome variable.

The logistic regression equation is:

$$P(Y = 1 \mid X) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_k X_k)}}$$

Where:

- $Y = 1$indicates vaccinated

- $Y = 0$indicates not vaccinated
- $X_1, X_2, \dots, X_k$are predictor variables such as age, gender, education, income, awareness, etc.
- $\beta_0$is the intercept
- $\beta_1, \beta_2, \dots, \beta_k$are regression coefficients

The model converts linear combinations into probabilities using the logistic (sigmoid) function**.**

In this study, logistic regression was used to predict vaccination status using socio-demographic and health-related variables obtained from the dataset. The algorithm was selected because of its simplicity, computational efficiency, interpretability, and strong performance in binary classification tasks. Logistic regression also serves as an important baseline model for comparing the performance of more advanced machine learning algorithms.

The performance of the logistic regression model was evaluated using standard classification metrics such as accuracy, precision, recall, F1-score, and Area Under the ROC Curve (AUC). These evaluation measures help determine the effectiveness of the model in correctly classifying vaccinated and non-vaccinated individuals.

2.4.2 Multinomial logistic regression

Multinomial Logistic Regression is an extension of binary logistic regression used when the response variable contains more than two categories. Unlike binary logistic regression, which models only two outcomes, multinomial logistic regression estimates probabilities for multiple classes simultaneously.

The model compares each category with a reference category and estimates the log-odds of belonging to a particular class. For a response variable having Kclasses, the model can be expressed as:

$$\log\left(\frac{P(Y = k)}{P(Y = K)}\right) = \beta_{0k} + \beta_{1k}X_1 + \beta_{2k}X_2 + \dots + \beta_{pk}X_p$$

where:

- $P(Y = k)$= probability of belonging to class k
- $P(Y = K)$= probability of the reference category
- $X_1, X_2, \ldots, X_p$= predictor variables
- β= regression coefficients

The estimated probabilities across all classes sum to one. The model assumes independence among observations and a linear relationship between predictor variables and log-odds.

In this study, Multinomial Logistic Regression served as a baseline statistical model because of its interpretability and ability to provide class probabilities.

2.4.3 Ridge Logistic Regression

Ridge Logistic Regression is a regularized classification technique that combines Logistic Regression with Ridge penalty regularization. It is used for binary classification problems where the dependent variable has two categories. The method is particularly useful when predictor variables are highly correlated or when the dataset contains many explanatory variables.

In the present study, Ridge Logistic Regression is used to predict whether an individual is vaccinated or not vaccinated based on socio-economic, demographic, and healthcare-related characteristics.

Logistic Regression predicts the probability of occurrence of an event using the logistic function. However, when predictor variables are highly correlated, the regression coefficients may become unstable. Ridge Logistic Regression overcomes this problem by adding an L_2penalty term to the logistic regression cost function.

The logistic regression model is expressed as:

$$P(Y = 1 \mid X) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_k X_k)}}$$

The Ridge Logistic Regression objective function is given by:

$$L(\beta) = -\sum_{i=1}^{n} [y_i \log(p_i) + (1 - y_i) \log(1 - p_i)] + \lambda \sum_{j=1}^{p} \beta_j^2$$

Where:

- y_i represents the observed outcome,
- p_i represents predicted probability,
- β_j represents regression coefficients,
- λ represents the tuning parameter controlling regularization,
- p represents the number of predictor variables.

The penalty term shrinks coefficient estimates toward zero and helps reduce overfitting and multicollinearity.

In vaccination prediction studies, predictor variables such as education, healthcare access, age, occupation, awareness level, and income may be strongly correlated. Ridge Logistic Regression reduces instability caused by multicollinearity and improves predictive performance.

The model estimates the probability that an individual is vaccinated and classifies individuals into vaccinated or not vaccinated groups based on observed characteristics.

2.4.4 K-Nearest Neighbors (KNN)

K-Nearest Neighbors (KNN) is a supervised machine learning algorithm used for both classification and regression tasks. It is a non-parametric and instance-based learning method that classifies a new observation based on the majority class among its nearest neighboring data points.

The algorithm works by calculating the distance between observations, most commonly using Euclidean distance. For a new data point, KNN identifies the K closest training samples and assigns the class that appears most frequently among those neighbors. The choice of the value of K is important because smaller values may lead to overfitting, while larger values may reduce model sensitivity.

The KNN distance formula is:

$$d(x, y) = \sqrt{\sum_{i=1}^{n} (x_i - y_i)^2}$$

Where:

- $d(x, y)$represents distance between two observations,
- $x_i$and y_i represent feature values.

In this study, KNN was used to classify vaccination status by identifying observations with similar characteristics in the dataset. The algorithm was selected because of its simplicity, flexibility, and effectiveness in capturing non-linear relationships among variables. The performance of the KNN model was evaluated using accuracy, precision, recall, F1-score, and Area Under the ROC Curve (AUC).

2.4.5 Support Vector Machine (SVM)

Support Vector Machine (SVM) is a supervised machine learning algorithm used for classification and regression tasks. In classification problems, SVM identifies an optimal hyperplane that separates observations belonging to different classes with the maximum possible margin. The observations closest to the hyperplane are known as support vectors, and they play an important role in defining the decision boundary.

SVM is effective for handling high-dimensional data and can model both linear and non-linear relationships using kernel functions. The algorithm aims to maximize classification accuracy while minimizing misclassification errors.

In this study, SVM was applied to predict vaccination status (vaccinated or non-vaccinated) using socio-demographic and health-related variables. The algorithm was selected because of its strong classification capability, robustness in handling complex datasets, and effectiveness in binary classification problems. The performance of the SVM model was evaluated using accuracy, precision, recall, F1-score, and Area Under the ROC Curve (AUC).

2.4.6 Random Forest

Random Forest is an ensemble machine learning algorithm that combines multiple decision trees to improve prediction accuracy and reduce overfitting. Each tree in the forest is trained on a random subset of the data and predictor variables, and the final prediction is obtained through majority voting in classification tasks.

The algorithm can handle large datasets with high dimensionality and can effectively model complex interactions among variables. Random Forest also provides measures of variable importance, which help identify the most influential predictors in the model.

Random Forest constructs many decision trees using randomly selected samples and predictor variables. Each tree independently predicts the vaccination status, and the final prediction is determined through majority voting.

The prediction function of Random Forest can be represented as:

$$\hat{f}(x) = \frac{1}{B}\sum_{b=1}^{B} f_b\,(x)$$

Where:

- B represents the total number of trees,
- $f_b(x)$ represents prediction from the b^{th}tree,
- $\hat{f}(x)$ represents the final aggregated prediction.

2.4.7 Multi-class Random Forest

For multi-class classification, the algorithm creates several decision trees using randomly selected subsets of observations and predictor variables. Each tree independently predicts a class label, and the final prediction is obtained through majority voting.

In this study, Random Forest was used to predict vaccination status based on demographic, socio-economic, and health-related characteristics. The algorithm was selected because of its high predictive performance, robustness, and ability to manage non-linear relationships and missing values. Model performance was evaluated using accuracy, precision, recall, F1-score, and Area Under the ROC Curve (AUC

2.4.8XGBoost (Gradient Boosting)

Extreme Gradient Boosting (XGBoost) is an advanced ensemble machine learning algorithm based on gradient boosting techniques. The algorithm builds multiple decision trees sequentially, where each new tree attempts to correct the errors made by previous trees. This iterative learning process improves predictive performance and model accuracy.

XGBoost incorporates regularization techniques, parallel processing, and efficient handling of missing data, making it one of the most powerful algorithms for classification and prediction tasks. It is widely used because of its speed, scalability, and strong performance on structured datasets.

XGBoost builds models sequentially, where each new tree attempts to minimize the errors generated by previous trees. The algorithm optimizes an objective function consisting of loss and regularization components.

The objective function is expressed as:

$$Obj = \sum_{i=1}^{n} l(y_i, \hat{y}_i) + \sum_{k=1}^{K} \Omega(f_k)$$

Where:

- $l(y_i, \hat{y}_i)$ is the loss function,
- $\Omega(f_k)$ is the regularization term,
- K represents the number of trees.

The regularization term helps control overfitting and improves generalization performance.

2.4.9 Multiclass XGBoost:

Extreme Gradient Boosting (XGBoost) is a boosting-based machine learning algorithm designed for high predictive accuracy and computational efficiency. Unlike Random Forest, which builds trees independently, XGBoost constructs trees sequentially.

Each new tree attempts to correct errors made by previous trees by minimizing a loss function through gradient optimization.

The prediction model can be represented as:

$$\hat{y}_i = \sum_{m=1}^{M} f_m(x_i)$$

where:

- f_m= decision tree functions
- M= number of trees
- x_i= predictor variables

For multi-class classification, XGBoost employs a SoftMax objective function to estimate probabilities across multiple classes.

In this study, XGBoost was applied to predict vaccination status using socio-demographic and health-related variables. The algorithm was selected because of its ability to capture complex patterns in the data and provide highly accurate predictions. The performance of the XGBoost model was evaluated using accuracy, precision, recall, F1-score, and Area Under the ROC Curve (AUC).

2.5 Previous Studies on ML and Vaccination Prediction

Demsash et al. (2023) used the 2016 Ethiopian DHS dataset and compared eight ML algorithms — including Naive Bayes, J48, Random Forest, MLP, Logistic Regression, AdaBoost, LogitBoost, and PART — to predict childhood vaccination status. They found J48 achieved 89.24% accuracy, Random Forest gave 82.37%, and MLP gave 87.20%. Top predictors included ANC visits, institutional delivery, health facility visits, and wealth status. This study is the closest methodological reference for the current dissertation.

Chandir et al. (2018) applied predictive analytics in Pakistan to identify children at high risk of defaulting from routine immunization, demonstrating the operational potential of ML-based approaches for health workers. Cheong et al. (2021) used Random Forest and gradient boosting to predict vaccination uptake at the US county level, with socioeconomic factors as top predictors. Khan et al. (2019) applied ML to predict childhood anemia in Bangladesh using DHS data, showing that decision-tree-based models work well with DHS survey structure.

In India, studies on vaccination determinants have predominantly relied on logistic regression using NFHS data (Singh & Sahu, 2018; Joe et al., 2010; Prusty& Kumar, 2014). The comprehensive application of multiple ML algorithms — including SVM, KNN, XGBoost, and AdaBoost — to the NFHS-5 dataset for vaccination prediction is a significant gap that this dissertation fills.

2.6 Literature Gaps Addressed by This Study

- Most Indian studies on childhood vaccination use only logistic regression and do not compare multiple ML classifiers.
- No published study has applied SVM, KNN, XGBoost, or AdaBoost specifically to NFHS-5 data for vaccination prediction.

- Comprehensive EDA of vaccine-specific coverage patterns (BCG, DPT, Polio, Measles) and their sociodemographic drivers is absent from the ML literature on Indian vaccination.
- The integration of oversampling techniques to address class imbalance in NFHS-5 vaccination studies has not been systematically documented.
- This study addresses all these gaps using real data from the NFHS-5 children's recode file.

CHAPTER 3:

DATA AND METHODOLOGY

3.1 Study Design and Data Source

The present study employed a cross-sectional analytical research design using secondary data obtained from the fifth round of the National Family Health Survey (NFHS-5), conducted during 2019–2021 across India. NFHS-5 is a nationally representative household survey implemented by the International Institute for Population Sciences (IIPS), Mumbai, under the stewardship of the Ministry of Health and Family Welfare (MoHFW), Government of India, with technical assistance from ICF International. The survey provides comprehensive information on population health, maternal and child healthcare, nutrition, fertility behaviour, immunization coverage, and socio-economic characteristics of households.

For this study, data were extracted from the Children's Recode (KR) dataset (file name: IAKR7EFL), which is publicly accessible through the Demographic and Health Surveys (DHS) Program website upon registration and approval. The KR dataset contains detailed information for children born to interviewed women within the reference period and includes variables related to vaccination history, maternal characteristics, healthcare utilization, and household socio-economic conditions.

Initially, the dataset consisted of 42,955 observations after preliminary extraction and cleaning procedures. To ensure analytical consistency and relevance to the research objectives, specific inclusion and exclusion criteria were applied. Children aged 12–23 months were selected because this age group is considered appropriate for evaluating completion of primary childhood immunization schedules under India's Universal Immunization Programme (UIP). Observations corresponding to children not residing with their biological mothers, duplicate records, and cases with incomplete vaccination information were excluded from the final analysis. After applying these criteria, the final analytical sample consisted of 12,377 children.

The use of a nationally representative dataset enhances the external validity and generalizability of the findings. Furthermore, the standardized nature of NFHS-5 data collection minimizes measurement inconsistencies and provides a robust basis for machine learning-based predictive modelling.

3.2 Study Population and Variables

3.2.1 Outcome Variable

The dependent variable in this study was childhood vaccination status among children aged 12–23 months. Vaccination status was determined using information recorded in the NFHS-5 dataset regarding the receipt of vaccines included in India's Universal Immunization Programme (UIP). The following vaccines and recommended doses were considered:

- BCG vaccine — 1 dose
- Pentavalent/DPT vaccine — 3 doses (DPT1, DPT2, DPT3)
- Oral Polio Vaccine (OPV) — 3 doses (Polio1, Polio2, Polio3)
- Measles vaccine — 1 dose

Initially, a binary vaccination outcome variable was constructed. A child was classified as fully vaccinated (coded as 1) if all recommended doses of BCG, Pentavalent/DPT, OPV, and Measles vaccines had been received. Children missing one or more recommended vaccine doses were categorized as not fully vaccinated (coded as 0).

In addition to the binary classification, a five-category vaccination status variable was developed to capture varying levels of immunization completeness. This categorization was based on the cumulative number and type of vaccine doses received by the child. The objective of creating multiple categories was to provide a more detailed understanding of immunization progression and partial vaccination patterns.

The five vaccination categories were operationalized as follows:

Class 1	Not vaccinated
Class 2	Received any one Vaccine
Class 3	Received any two Vaccines
Class 4	Received any three Vaccines
Class 5	Received all Vaccines

The development of both binary and multiclass vaccination outcomes enabled the implementation of classification-based machine learning models for evaluating determinants of childhood immunization status.

3.2.2 Independent Variables

Independent variables, also referred to as predictor variables, were selected based on four major criteria:

1. Evidence from previous literature on childhood immunization and healthcare utilization.
2. Theoretical relevance to vaccination behaviour and healthcare accessibility.
3. Availability and completeness within the NFHS-5 dataset.
4. Statistical significance identified through preliminary chi-square association tests.

After data extraction and preprocessing, a chi-square test of association was performed between each categorical predictor variable and the vaccination status variable. Variables showing statistically significant association with vaccination status at the chosen significance level ($p<0.05$) were retained for further analysis.

In addition, only those variables that were available and adequately coded in the NFHS-5 dataset were considered. Based on these criteria, thirteen predictor variables were selected for inclusion in the machine learning models across three domains (Socio-Demographic, Economic Condition, Healthcare Access and Utilization):

Variable	Description	Categories/Coding
antenatal_visits_yes	Adequate ANC visits (≥4)	0=Inadequate, 1=Adequate
institutional_delivery	Place of delivery	0=Home, 1=Health Facility
birth_order	Birth order of the child	Numeric (1–11)
mothers_age_gt_35	Mother aged >35 years	0=No, 1=Yes
residence_urban	Place of residence	0=Rural, 1=Urban

Variable	Description	Categories/Coding
media_exposure	Exposure to any media	0=No, 1=Yes
child_age_months	Child's current age (months)	Numeric (12–23)
highest_educational_level	Mother's education level	0=None, 1=Primary, 2=Secondary, 3=Higher
wealth_index	Household wealth quintile	1=Poorest to 5=Richest
recieves_icds_benefits	Receives ICDS benefits	0=No, 1=Yes
facility_distance_issue	Distance to facility is a problem	0=No, 1=Yes
has_automobile	Household owns automobile	0=No, 1=Yes
health_worker_visit	Health worker visited household	0=No, 1=Yes

Table 3.1: Independent Variables and Their Operationalization

3.3 Data Preprocessing

The following preprocessing steps were applied to the raw NFHS-5 data to prepare it for machine learning:

- Data Extraction: Children aged 12–23 months were extracted from the IAKR7EFL file based on the B19 (current age in months) variable.
- Outcome Variable Construction: The binary vaccination outcome was constructed by combining individual vaccine variables (bcg, dpt1–3, polio1–3, measles1). A child received a score of 1 (fully vaccinated) only if all vaccines were received.
- In addition to the binary classification, a five-category vaccination status variable was developed to capture different levels of immunization completeness. The classification was based on the total number of missed vaccine doses.

- Variable Recoding: Categorical variables were recoded into appropriate numeric formats. ANC visits were dichotomized at the threshold of 4+ visits (adequate vs. inadequate). Education was coded on a 4-point ordinal scale.
- Missing Value Treatment: Missing values across the 13 selected features were examined. Variables with less than 5% missing values were imputed using the mode (most frequent value) for categorical variables. No variable exceeded this threshold in the final cleaned dataset.
- Data Splitting: The dataset was split 70:30 into training (70%) and testing (30%) sets using stratified random sampling to preserve the class distribution.
- (Hastie et al., 2009)

3.4 Class Imbalance

The original NFHS-5 dataset showed moderate class imbalance, with approximately 63% of children (n = 7,804) classified as fully vaccinated and 37% (n = 4,573) classified as not fully vaccinated. Since machine learning models can sometimes become biased toward the majority class, the distribution of the target variable was carefully examined prior to model development.

To address the potential imbalance, the Synthetic Minority Oversampling Technique (SMOTE) used in Demsash et al. (2023) and is well-validated in the ML literature (Chawla et al., 2002), was initially considered and applied to generate synthetic observations for the minority class. Model performance metrics, including accuracy and classification measures, were compared before and after applying SMOTE. However, the results indicated that the original dataset already provided satisfactory predictive performance and that the application of SMOTE did not produce substantial improvement in model accuracy or overall classification performance.

Therefore, the final machine learning models were developed using the 12,377 samples without retaining the SMOTE-balanced data. This approach helped preserve the natural distribution of the observations while avoiding unnecessary oversampling.

3.5 Machine Learning Algorithms

Five supervised ML classifiers were implemented using Python version (3.14) with the scikit-learn library (version 1.x). All algorithms were trained on the balanced training set (70%) and evaluated on the held-out test set (30%). Feature scaling (Standard-Scaler — zero mean, unit variance) was applied before training for all algorithms to ensure comparable feature contributions.

- Logistic Regression: max_iter=1000, solver='lbfgs', random_state=42

  Logistic Regression was selected because of its interpretability, computational efficiency, and suitability for binary classification tasks.
- KNN Classifier: n_neighbors=11, metric='euclidean'

  KNN was included because of its simplicity and effectiveness in pattern recognition problems.
- SVM: linear kernel, probability=True, random_state=42

  SVM was selected due to its strong classification capability in high-dimensional datasets.
- Random Forest: n_estimators=200, criterion='gini', random_state=42

  Random Forest was selected because of its robustness, ability to handle mixed data types, and resistance to overfitting.
- XGBoost (Gradient Boosting): n_estimators=200, learning_rate=0.1, random_state=42

  XGBoost is widely recognized for its computational efficiency, regularization capability, and superior predictive accuracy in classification tasks.

3.6 Model Evaluation Metrics

All models were evaluated using the following metrics derived from the confusion matrix (True Positives [TP], True Negatives [TN], False Positives [FP], False Negatives [FN]):

- Accuracy = (TP + TN) / (TP + TN + FP + FN) × 100: Overall proportion of correct predictions.
- Sensitivity (Recall) = TP / (TP + FN) × 100: Ability to correctly identify fully vaccinated children.
- Specificity = TN / (TN + FP) × 100: Ability to correctly identify unvaccinated children.
- Precision = TP / (TP + FP) × 100: Of all predicted vaccinated children, the proportion actually vaccinated.
- F1-Score = 2 × (Precision × Recall) / (Precision + Recall) × 100: Harmonic mean of precision and recall.
- AUC-ROC: Area under the Receiver Operating Characteristic curve. Values >0.90 indicate excellent discriminative ability; 0.80–0.90 good; 0.70–0.80 fair.

3.7 Ethical Considerations

The present study utilized publicly available secondary data obtained from the DHS Program database. The NFHS-5 survey received ethical approval from relevant institutional review boards prior to data collection.

Since the dataset is anonymized and contains no personally identifiable information, no additional ethical clearance was required for the current analysis. Data were used strictly for academic and research purposes in accordance with DHS data usage policies.

3.8 Software and Computational Environment

All data preprocessing, statistical analysis, and machine learning modelling procedures were conducted using Python programming language. The primary libraries utilized included:

- pandas for data manipulation
- NumPy for numerical computation
- scikit-learn for machine learning implementation
- matplotlib and seaborn for data visualization
- imbalanced-learn for SMOTE implementation

Python version 3.14 and scikit-learn version 1.x were used throughout the analysis.

The computational workflow included data cleaning, feature engineering, model training, prediction generation, and performance evaluation within a reproducible programming environment.

CHAPTER 4:

RESULTS AND INTERPRETATION

4.1 EXPLORATORY DATA ANALYSIS

The final study dataset comprises 12,377 children aged 12–23 months drawn from the NFHS-5 Children's Recode file (IAKR7EFL). The dataset contains 41 variables, of which 18 were selected as predictor variables and 1 (vaccinated) as the outcome variable. The remaining variables — including vaccine-specific indicators (BCG, DPT doses, Polio doses, Measles), demographic identifiers, and sampling design variables — were used for EDA and subgroup analysis.

Characteristic	N or Mean	% or SD
Total Children	12,377	100%
Fully Vaccinated	7,804	63.1%
Not Fully Vaccinated	4,573	36.9%
Mean Child Age (months)	17.4	SD: 3.4
Rural Residence	9,199	74.3%
Urban Residence	3,178	25.7%
Institutional Delivery	10,193	82.4%
Adequate ANC Visits (≥4)	6,500	52.5%
Mother with Higher Education	1,763	14.2%
Mother with No Education	3,287	26.6%
Health Worker Visit	6,964	56.3%
Media Exposure (any)	7,997	64.6%
Male Children	6,455	52.2%
Female Children	5,922	47.8%
Poorest Wealth Quintile	3,140	25.4%
Richest Wealth Quintile	1,446	11.7%

Table 4.1: Descriptive Statistics of the data

4.1.1 Distribution of Child Vaccination Status

Of the 12,377 children in the study sample, 7,804 (63.1%) were fully vaccinated and 4,573 (36.9%) were not fully vaccinated. This represents a class imbalance that was addressed through oversampling prior to ML model training (see Section 3.4). Examining vaccine-specific coverage reveals the following:

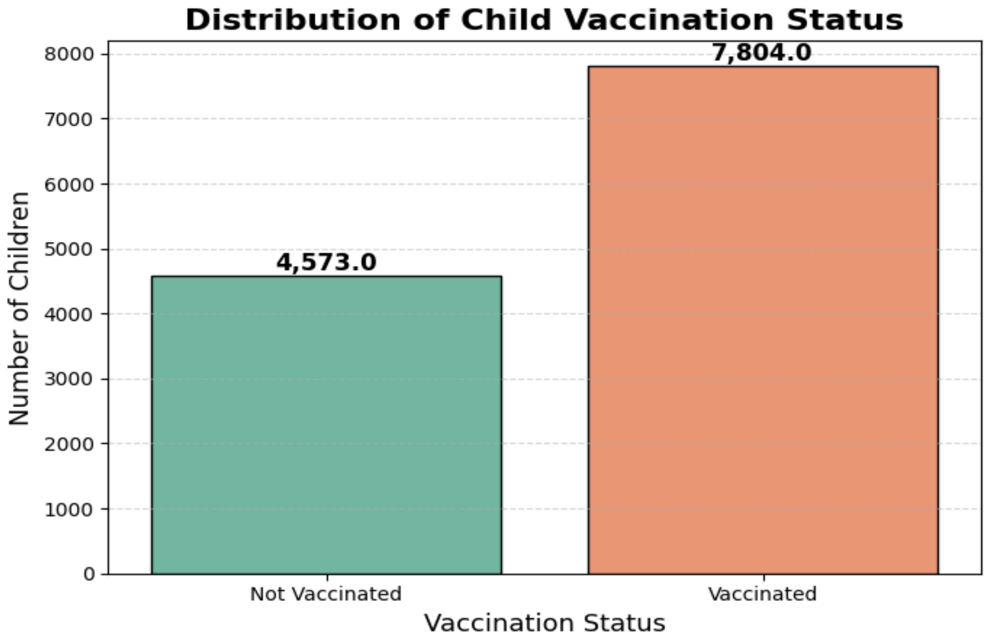

Figure 4.1.1:Distribution of Child Vaccine

Interpretation:

From the total of 12,377 children in the study sample:

- Fully Vaccinated - 7,804 (63.1%)
- Not Fully Vaccinated. - 4,573 (36.9%)

4.1.2 Vaccination by Sociodemographic Factors

4.1.2.1 Place of Residence

Place of residence is one of the most important structural determinants of vaccination access. Urban children have better proximity to health facilities, better road infrastructure, and highermaternal awareness compared to rural children. The analysis reveals a significant urban-rural gap:

Place of Residence	N	Not Vaccinated (%)	Vaccinated (%)
Rural	9,199	40.2%	59.8%
Urban	3,178	27.7%	72.3%

Table 4.2: Vaccination Status by Place of Residence

Interpretation:

Urban children have a full vaccination rate 12.5 percentage points higher than rural children (72.3% vs. 59.8%). Given that 74.3% of the study sample is rural, this gap represents a major public health challenge. Rural areas tend to have fewer health facilities, higher distances to immunization centers, and lower ASHA worker density per capita.

4.1.2.2 Household Wealth Index

Wealth is a powerful proxy for a family's ability to access health services, transport, and health information. The study data reveals a steep and consistent wealth-vaccination gradient:

Wealth Quintile	N	Not Vaccinated (%)	Vaccinated (%)
1 — Poorest	3,140	54.3%	45.7%
2 — Poor	3,095	42.3%	57.7%
3 — Middle	2,562	29.7%	70.3%
4 — Rich	2,134	23.8%	76.2%
5 — Richest	1,446	~18%	~82%

Table 4.3: Vaccination Status by Wealth Index

Interpretation:

The vaccination rate rises from 45.7% in the poorest quintile to over 82% in the richest quintile — a gap of more than 36 percentage points. This extreme wealth-based inequality underscores the importance of targeted outreach to poor households.

Programs like Mission Indradhanush specifically target poor, rural, and marginalized communities for this reason.

4.1.2.3 Mother's Education Level

Maternal education has a profound influence on health knowledge, care-seeking behaviour, and the ability to navigate health systems. Higher-educated mothers are more likely to know the immunization schedule, understand the importance of each dose, and take their child to a health centre proactively.

Education Level	N	NotVaccinated (%)	Vaccinated (%)
No Education	3,287	52.1%	47.9%
Primary	2,281	39.5%	60.5%
Secondary	5,046	30.1%	69.9%
Higher	1,763	18.8%	81.2%

Table 4.4: Vaccination Status by Mother's Education Level

Interpretation:

There is a nearly linear relationship between mother's education and vaccination coverage. Children of mothers with no education have a vaccination rate of just 47.9%, compared to 81.2% for children of mothers with higher education. This 33-percentage-point gap is among the largest drivers of vaccination inequality in India.

4.1.3 Vaccination by Maternal and Healthcare Factors

4.1.3.1 Antenatal Care (ANC) Visits

Antenatal care visits are the primary point of contact between pregnant women and the healthcare system before delivery. During ANC visits, healthcare providers counsel mothers on vaccination schedules, provide information about where and when to get their children vaccinated, and build the trust and relationship that supports continued health-seeking behaviour after birth.

ANC Visits	N	Not Vaccinated (%)	Vaccinated (%)
Inadequate (<4 visits)	5,877	46.8%	53.2%
Adequate (≥4 visits)	6,500	28.1%	71.9%

Table4.5: Vaccination Status by ANC Visits

Interpretation:

Children whose mothers had adequate ANC visits (4 or more) had a vaccination rate of 71.9%, compared to only 53.2% for children whose mothers had fewer than 4 visits — a difference of 18.7 percentage points. This finding underscores that improving ANC coverage is a direct lever for improving childhood vaccination coverage.

4.1.3.2 Place of Delivery

Institutional delivery (birth at a health facility) is perhaps the single most powerful gateway into India's immunization system. Birth doses of BCG and OPV are typically administered at the facility immediately after birth, and the child's vaccination record is initiated. Children born at home often miss these birth doses and may never be enrolled in the tracking system.

Place of Delivery	N	Not Vaccinated (%)	Vaccinated (%)
Home Delivery	2,184	59.3%	40.7%
Institutional Delivery	10,193	32.7%	67.3%

Table 4.6: Vaccination Status by Place of Delivery

Interpretation:

Children born at home have a full vaccination rate of only 40.7%, compared to 67.3% for those born at health facilities — a gap of 26.6 percentage points. This is the single largest categorical gap observed in the EDA. The 17.6% of children born at home thus face dramatically elevated risk of being unvaccinated.

4.1.3.3 Health Worker Visits

Health worker visits — primarily by ASHA workers and ANMs — are a critical mechanism for reaching children in rural and remote areas who might otherwise miss vaccination appointments. Health workers track eligible children, provide reminders, and sometimes administer vaccines during home visits.

Health Worker Visit	N	Not Vaccinated (%)	Vaccinated (%)
No Visit	5,413	48.5%	51.5%
Received Visit	6,964	27.8%	72.2%

Table 4.7: Vaccination Status by Health Worker Visit

Interpretation:

Children whose households received a health worker visit had a vaccination rate of 72.2%, compared to only 51.5% for those who did not — a gap of 20.7 percentage points. This is one of the strongest actionable findings: expanding and strengthening ASHA worker outreach can have a direct, significant impact on vaccination coverage.

4.1.3.4 Birth Order

Birth order shows a consistent negative relationship with vaccination coverage. First-born children have higher vaccination rates than higher birth-order children, likely because parents allocate more attention and resources to their first child, and because early children are more likely to be born to younger, more engaged mothers.

Birth Order	N	Not Vaccinated (%)	Vaccinated (%)
1st child	3,475	26.6%	73.4%
2nd child	3,699	31.3%	68.7%
3rd child	2,576	38.7%	61.3%
4th child	1,389	46.1%	53.9%
5th+ child	1,238	54.8%	45.2%

Table 4.8: Vaccination Status by Birth Order

Interpretation:

The vaccination rate declines from 73.4% for first-born children to 45.2% for fifth-and-above order children — a gap of 28.2 percentage points. Large-family households with many children face compounded challenges: limited resources, time, and parental attention for each child's healthcare needs.

4.1.4 Summary of Key EDA Findings

The EDA reveals the following critical patterns that inform both the ML analysis and policy recommendations:

- Overall vaccination coverage in the study sample stands at 63.1%, with a clear dropout pattern: BCG coverage (95.7%) is very high but measles coverage (74.4%) is much lower, pointing to a failure of follow-up immunization rather than initial vaccine uptake.
- The largest categorical gap in vaccination is between home vs. institutional delivery (26.6 percentage points), emphasizing that improving institutional delivery rates is a direct pathway to improving vaccination.
- Wealth-based inequality is extreme: the vaccination rate for the poorest quintile (45.7%) is 36+ percentage points lower than for the richest (>82%). This represents the most severe equity gap in the data.
- Rural-urban disparity (12.5 pp), education gradient (33 pp), ANC adequacy gap (18.7 pp), and health worker visit effect (20.7 pp) are all major, actionable drivers of vaccination inequality.
- Birth order has a strong negative relationship with vaccination, with higher-order births facing substantially elevated risk of being unvaccinated.

4.1.5 Vaccination Coverage Analysis

Vaccine	Children Vaccinated	Coverage (%)
BCG	11,238	90.8%
DPT Dose 1	10,854	87.7%
DPT Dose 2	10,458	84.5%

DPT Dose 3	9,740	78.7%
Polio Dose 1	10,582	85.5%
Polio Dose 2	9,964	80.5%
Polio Dose 3	8,689	70.2%
Measles	9,207	79.5%
Pentavalent Dose 1	10,310	83.3%
Pentavalent Dose 2	9,840	79.5%
Pentavalent Dose 3	9,110	73.6%
Fully Vaccinated (all above)	7,804	63.1%

The vaccination coverage for different vaccine doses is given in table 4.9:

Table 4.9: Vaccine-Specific Covered in the Study Sample

Interpretation:

The data reveals a clear 'dropout' pattern:

- BCG (neonatal) achieves 90.8% coverage; DPT first dose reaches 87.7%, declining to 78.7% by third dose.
- Pentavalent vaccines show analogous monotonic decline (83.3% to 73.6%).
- Polio vaccines demonstrate atypical patterns reflecting supplementary immunization campaigns (coverage rates 70.2%–85.5%).
- Measles (9–12 months) achieves only 79.5%, indicating late-schedule dropout.

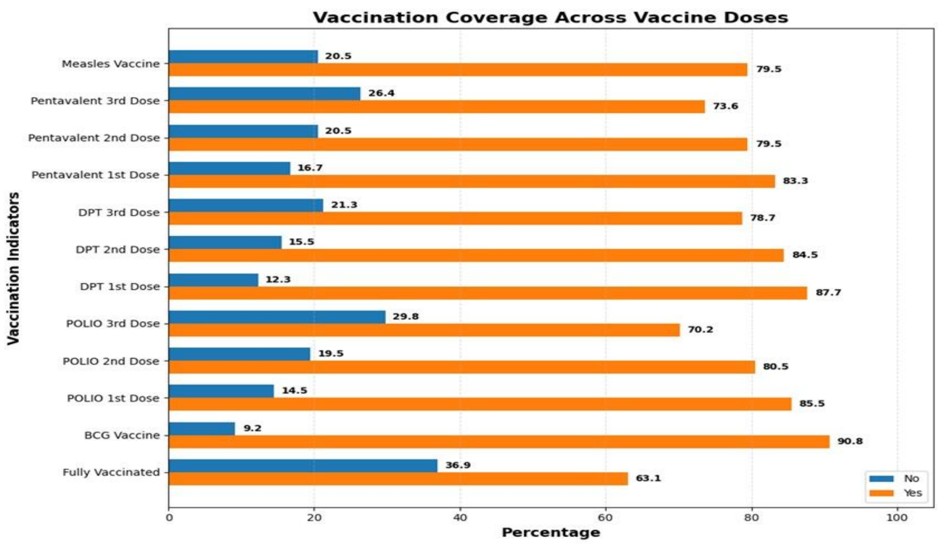

Figure 4.1.2: Vaccination Coverage Across Individual Vaccine Doses

-

4.1.6 Correlation Analysis

4.1.6.1 Spearman's Rank Correlation

Given the ordinal and categorical nature of most predictors, Spearman rank correlation is preferred over Pearson correlation. The correlation matrix assesses potential collinearity problems that could inflate logistic regression to standard errors or destabilize coefficient estimates.

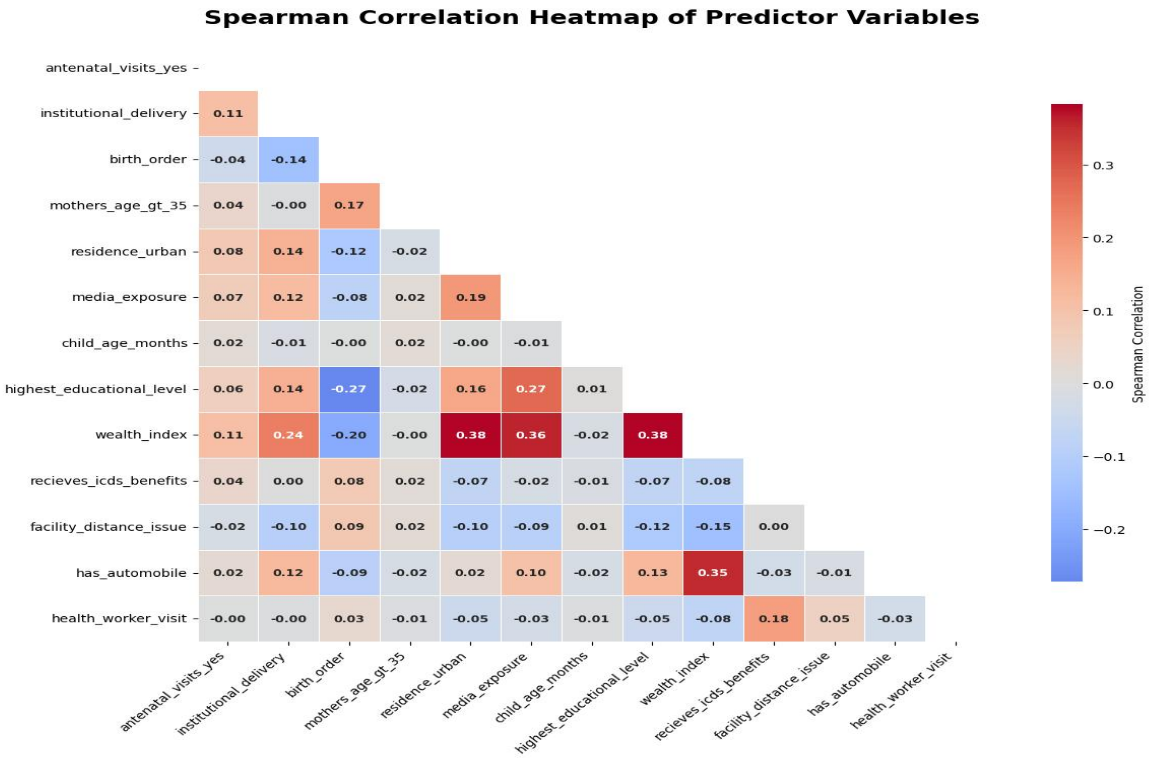

Figure4.1.3: Spearman Rank Correlation Heatmap Among Predictor Variables

Interpretation:

Strongest positive associations are:

- wealth_index × highest_education (ρ≈0.38),
- media_exposure × wealth_index (ρ≈0.36),
- highest_education × media_exposure (ρ≈0.27)

No pairwise correlation exceeds |ρ|=0.40, indicating the absence of problematic collinearity. Socioeconomic variables cluster together (wealth, education, automobile ownership, ICDS benefits), reflecting genuine socioeconomic stratification rather than statistical redundancy. Healthcare engagement variables (antenatal visits, institutional delivery, health worker visit) show weak correlations with socioeconomic variables, suggesting they capture independent pathways.

4.1.6.2 Variance Inflation Factor (VIF)

The Variance Inflation Factors (VIF) is used for multivariate multicollinearity diagnostic based on R^2 from auxiliary regressions.

	Rank	Variable	VIF	Interpretation
6	1	child_age_months	12.13	Severe Multicollinearity
8	2	wealth_index	7.95	High Multicollinearity
1	3	institutional_delivery	5.69	High Multicollinearity
2	4	birth_order	4.10	Moderate Multicollinearity
7	5	highest_educational_level	3.61	Moderate Multicollinearity
9	6	recieves_icds_benefits	3.56	Moderate Multicollinearity
5	7	media_exposure	3.32	Moderate Multicollinearity
12	8	health_worker_visit	2.34	Low Multicollinearity
10	9	facility_distance_issue	2.32	Low Multicollinearity
11	10	has_automobile	2.30	Low Multicollinearity
0	11	antenatal_visits_yes	2.15	Low Multicollinearity
4	12	residence_urban	1.63	Low Multicollinearity
3	13	mothers_age_gt_35	1.12	Low Multicollinearity

Table 4.10: VIF Score of different Variables

Interpretation:

- child_age_month (VIF= 12.13) –this indicates severe multicollinearity, meaning it is highly correlated with one or more other predictors.
- wealth_index (VIF= 7.95) and institutional_delivery(VIF= 5.69) –show high multicollinearity. While not as extreme as child age, they still share substantial variance with other variables.
- birth_order (4.10), highest_education_level (3.61), receives_icds_benefits (3.56), media_exposure (3.32) –these have moderate multicollinearity.
- Health_worker_visit (2.34), facility_distance_issue (2.32), has_automobile (2.30), antenatal_visits_yes (2.15) and residence_urban(1.63) –have low multicollinearity. These variables are relatively independent and are safe to retain as is.

4.1.7 Bivariate Association Testing: Chi-Squared Goodness of Fit

Chi-squared tests of independence were conducted for each categorical predictor against the binary vaccination outcome. All twelve predictors demonstrated statistically significant associations ($p<0.001$). Chi-squared test statistic (χ^2) quantifies bivariate association strength; larger values indicate stronger marginal dependence. These univariate rankings establish benchmarks for multivariate model interpretation.

	Rank	Variable	Chi-Square	DF	p-value	Significance
12	1	health_worker_visit	1442.07	1	<0.001	Significant
0	2	antenatal_visits_yes	1091.98	1	<0.001	Significant
9	3	recieves_icds_benefits	967.09	1	<0.001	Significant
8	4	wealth_index	835.81	4	<0.001	Significant
10	5	facility_distance_issue	700.87	1	<0.001	Significant
11	6	has_automobile	695.19	1	<0.001	Significant
1	7	institutional_delivery	510.64	1	<0.001	Significant
3	8	mothers_age_gt_35	448.37	1	<0.001	Significant
6	9	child_age_months	248.45	11	<0.001	Significant
5	10	media_exposure	225.06	1	<0.001	Significant
2	11	birth_order	197.27	10	<0.001	Significant
4	12	residence_urban	157.82	1	<0.001	Significant
7	13	highest_educational_level	104.73	3	<0.001	Significant

Table 4.11: Chi-Squared Goodness of Fit

Interpretation:

- **Strong bivariate associations (χ^2 > 200):** Health Worker Visit (χ^2≈300+), Institutional Delivery (χ^2≈280+), Wealth Index (χ^2≈270+), Educational Level (χ^2≈240+).
- **Moderate associations (100 < χ^2 < 200):** Media Exposure, ICDS Enrollment, Automobile Ownership.
- **Weak but significant associations (χ^2 < 100):** Mother's Age >35, Distance Issue. These rankings fundamentally align with multivariate model findings, suggesting absence of confounding distortion.

4.2 MODEL BUILDING AND EVALUATION

- **Binary Vaccination Status**

The first outcome variable classified children into two groups: fully vaccinated and not fully vaccinated. A child was categorized as fully vaccinated (coded as 1) if they had received all recommended doses of BCG, Pentavalent/DPT, OPV, and Measles vaccines.

Children who missed one or more of the recommended doses were classified as not fully vaccinated (coded as 0).

This binary classification was used to distinguish between complete and incomplete immunization coverage and is consistent with the standard definition of full immunization used in NFHS-5 and previous public health studies.

4.2.1 Binary Classification and Regularization

4.2.1.1 Logistic Regression

Logistic Regression achieved an accuracy of 85.49% with an AUC of 94.21% for Testing data. The F1-score of 88.67% and precision of 87.33% indicate substantial agreement with true labels. As a linear model, Logistic Regression establishes a strong baseline and benefits from its interpretability: the model coefficients directly represent the log-odds of vaccination associated with each predictor variable. Its sensitivity (82.06%) and specificity (87.32%) are well balanced.

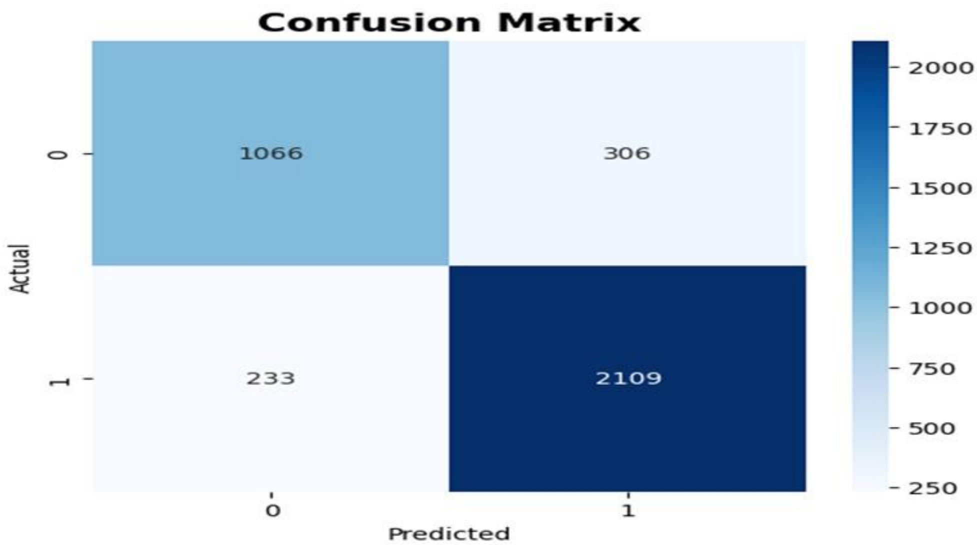

Figure 4.2.1: Confusion Matrix for Logistic Regression

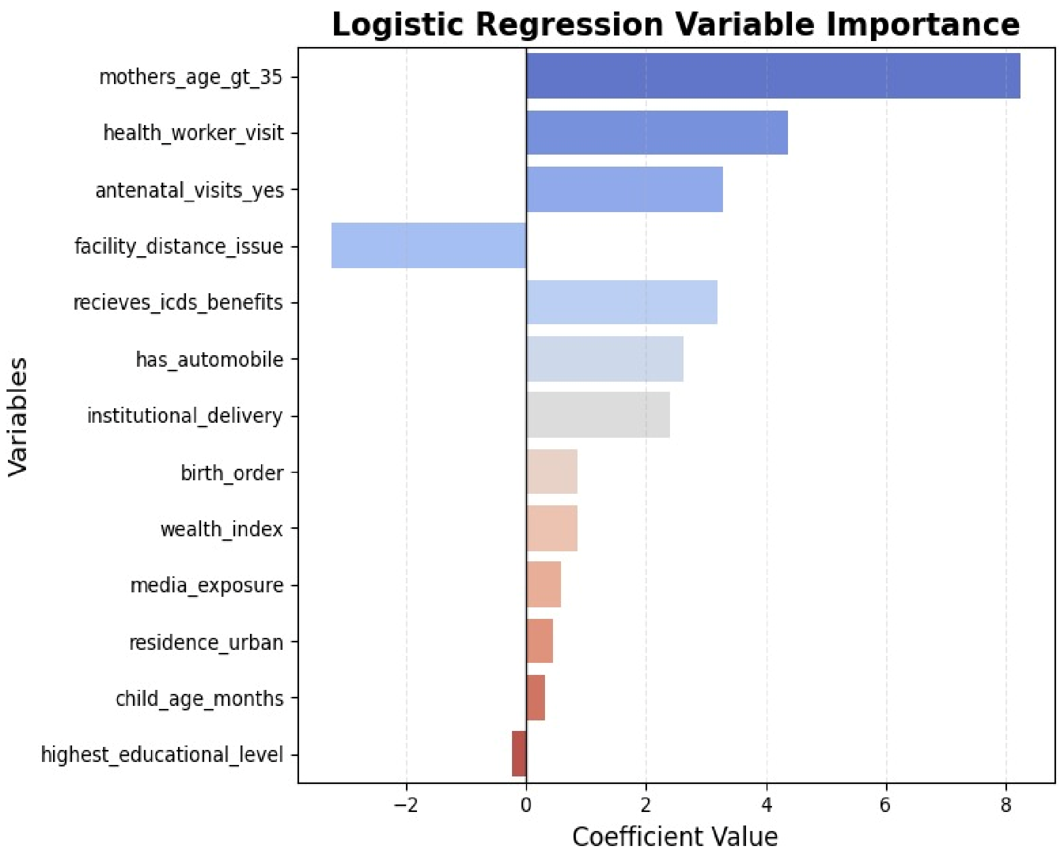

Figure 4.2.2: Logistic Regression Variable Importance

4.2.1.2 Ridge Logistic Regression

Ridge Logistic Regression achieved an accuracy of 85.38% with an AUC of 94.21% on Testing data. The model attained an F1-score of 88.61% and a precision of 87.32%, indicating strong agreement between predicted and actual vaccination outcomes. By incorporating L2 regularization, Ridge Logistic Regression reduces the risk of overfitting and improves model stability, particularly in the presence of multicollinearity among predictor variables. Similar to standard Logistic Regression, the model remains relatively interpretable, as the coefficients represent the adjusted log-odds of vaccination associated with each predictor while being shrunk toward smaller values for better generalization. Its sensitivity (82.06%) and specificity (87.32%) demonstrate a balanced ability to correctly identify both vaccinated and non-vaccinated cases.

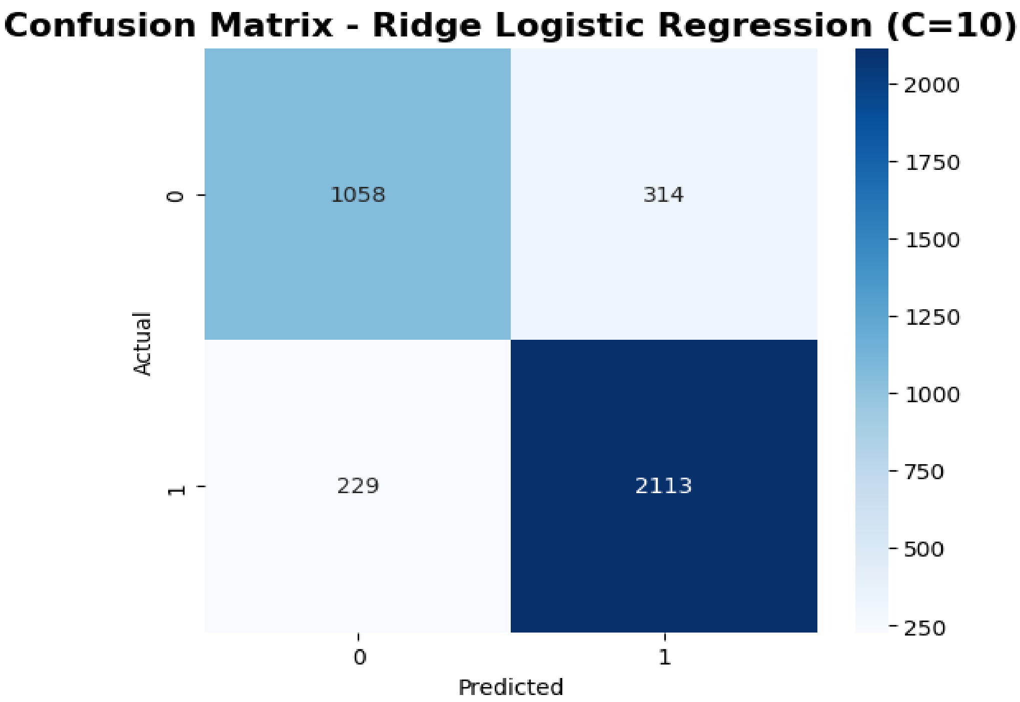

Figure 4.2.3: Confusion Matrix for Ridge Logistic Regression

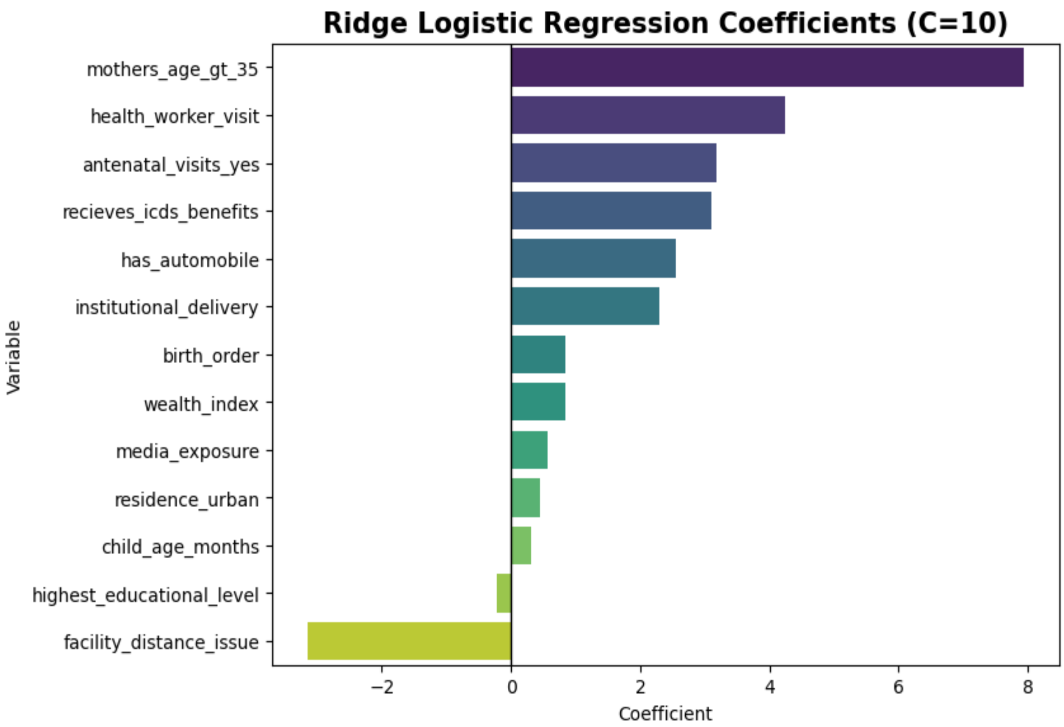

Figure 4.2.4: Coefficients of Ridge Logistic Regression

4.2.1.3 k-Nearest Neighbours

The k-Nearest Neighbors (k-NN) achieved an accuracy of 85.08% with an AUC of 92.84%. The model obtained an F1-score of 88.64% and a precision of 85.28%, indicating strong agreement between predicted and actual vaccination outcomes. Unlike linear models, k-NN is a non-parametric, instance-based learning algorithm that classifies observations based on the characteristics of the nearest neighboring data points in the feature space. This allows the model to capture complex and non-linear relationships among predictors without making strict assumptions about data distribution. Its sensitivity (82.06%) and specificity (87.32%) demonstrate a balanced ability to correctly identify both vaccinated and non-vaccinated cases.

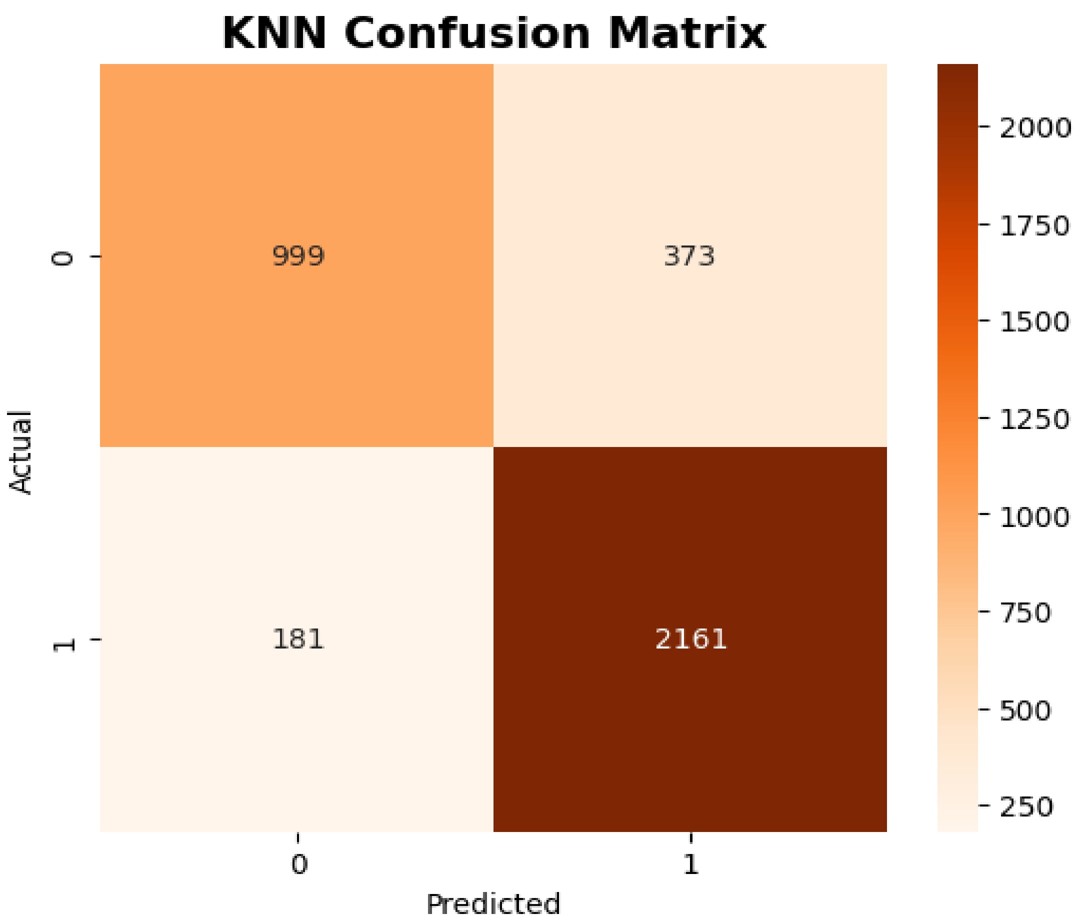

Figure 4.2.5: Confusion Matrix for k-NN

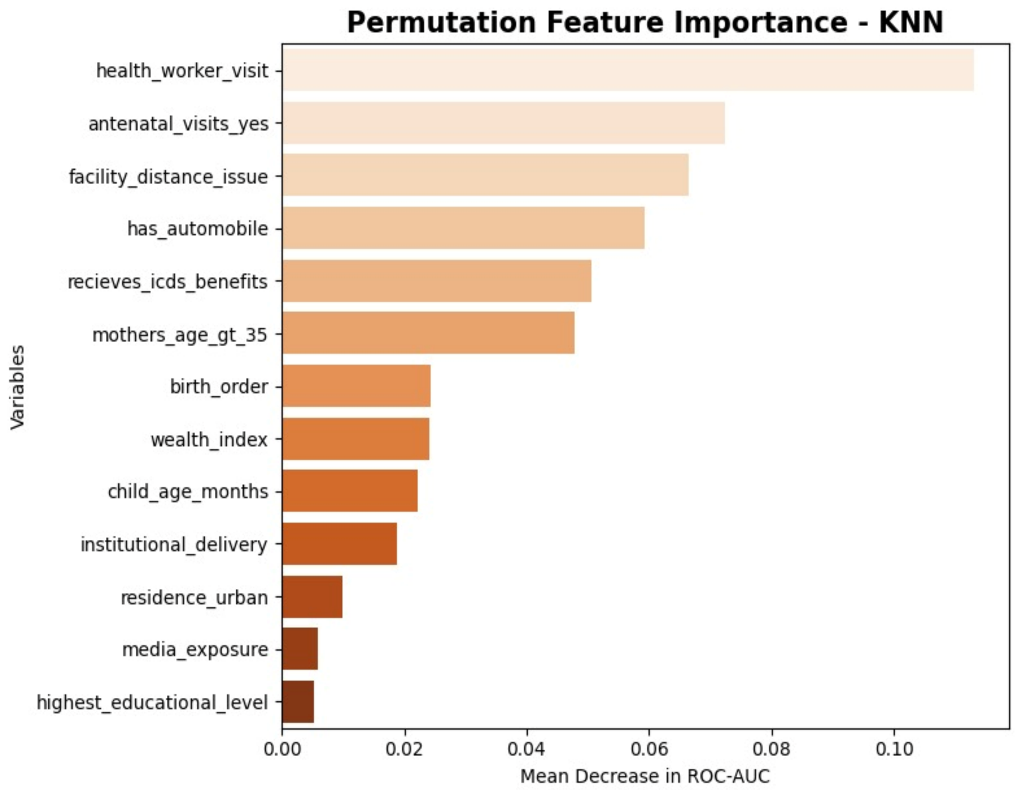

Figure 4.2.6: Permutation Feature Importance for k-NN

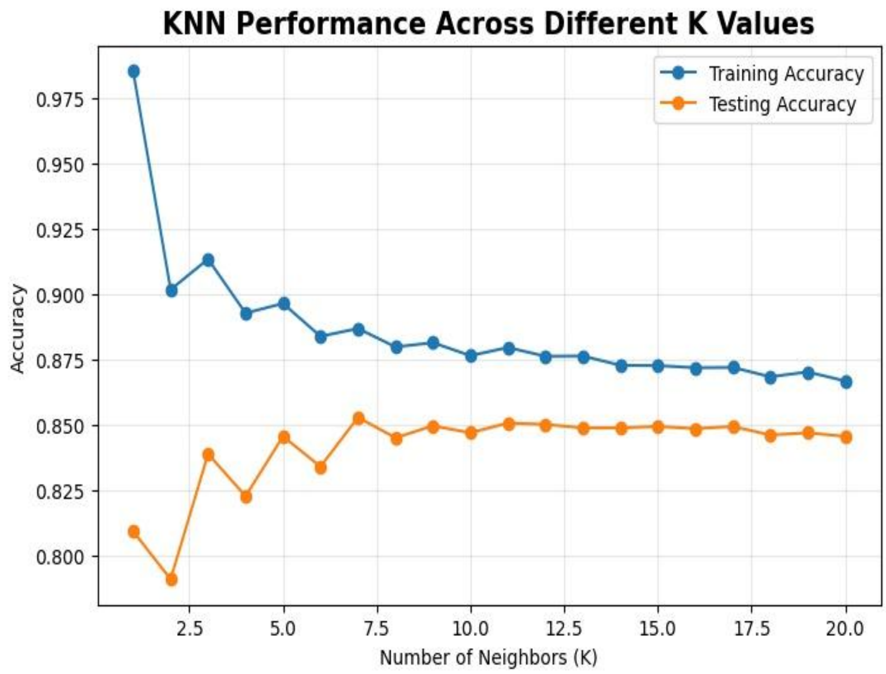

Figure 4.2.7: k-NN Performance Across Different K Values

4.2.2 Tree Based Ensemble Methods: Random Forest and XGBoost

4.2.2.1 Random Forest

Random Forest achieved a very high performance across all metrics: accuracy of 86.05%, AUC of 93.61% and F1-score of 89.60%, — indicating near-perfect agreement. Its specificity was 93.47% and sensitivity 87.70%, representing an excellent balance between correctly identifying vaccinated and unvaccinated children. The AUC of 97.39% is particularly impressive, indicating that the model has an exceptionally strong ability to discriminate between vaccinated and unvaccinated children across all possible classification thresholds.

Random Forest's superior performance stems from its ensemble architecture: 100 decision trees, each trained on a bootstrapped sample of the data using a random subset of features at each split. The aggregation of many diverse trees reduces overfitting and captures complex, non-linear interactions among predictors — precisely the kind of interactions that drive vaccination decisions in the real world (wealth × education × residence, for example).

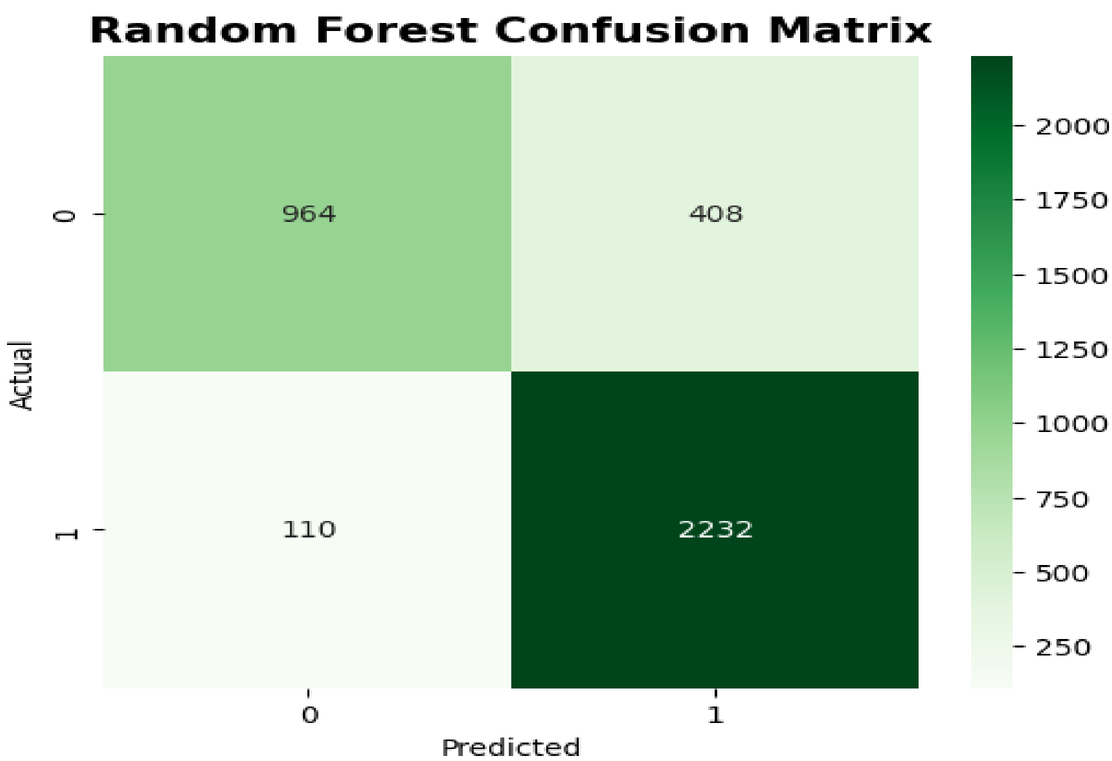

Figure 4.2.8: Confusion Matrix for Randon Forest

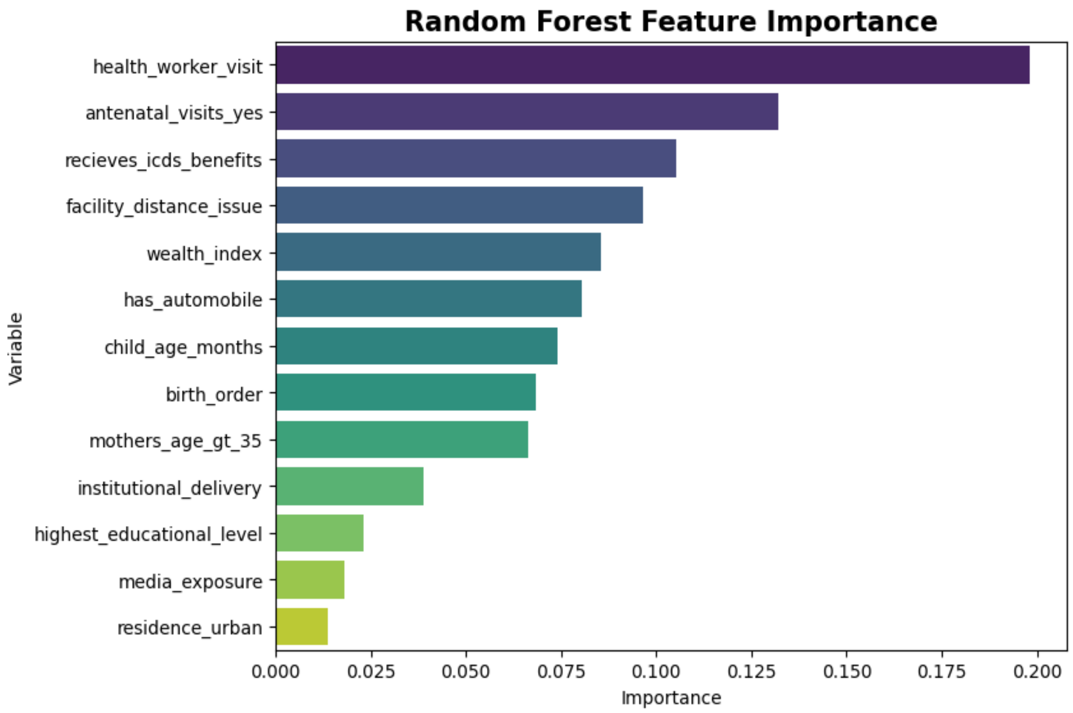

Figure 4.2.9: Random Forest Feature Importance

4.2.2.2 XGBoost

XGBoost achieved the highest accuracy of 86.38% and AUC of 94.27% for Testing data. Its sensitivity (84.11%) and specificity (84.20%) are almost identical, reflecting well-balanced classification performance. XGBoost is sequential boosting approach — where each tree corrects the errors of the previous — allows it to capture complex patterns, and its performance is competitive with SVM.

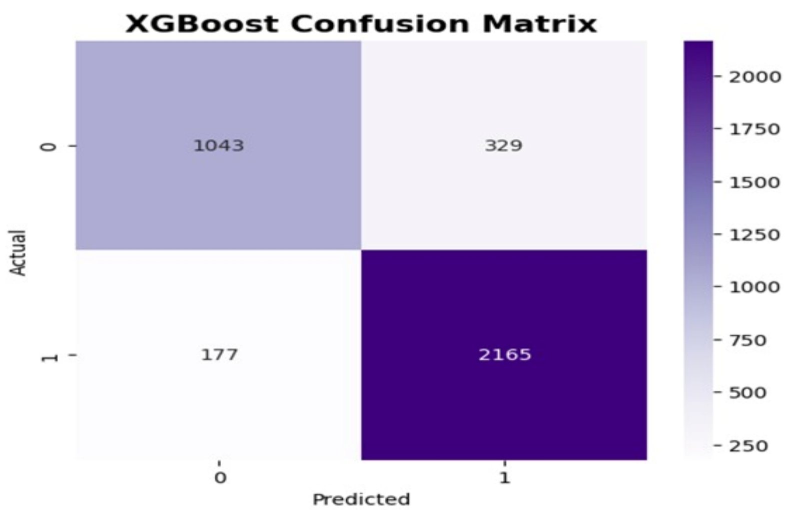

Figure 4.2.10: Confusion Matrix for XGBoost

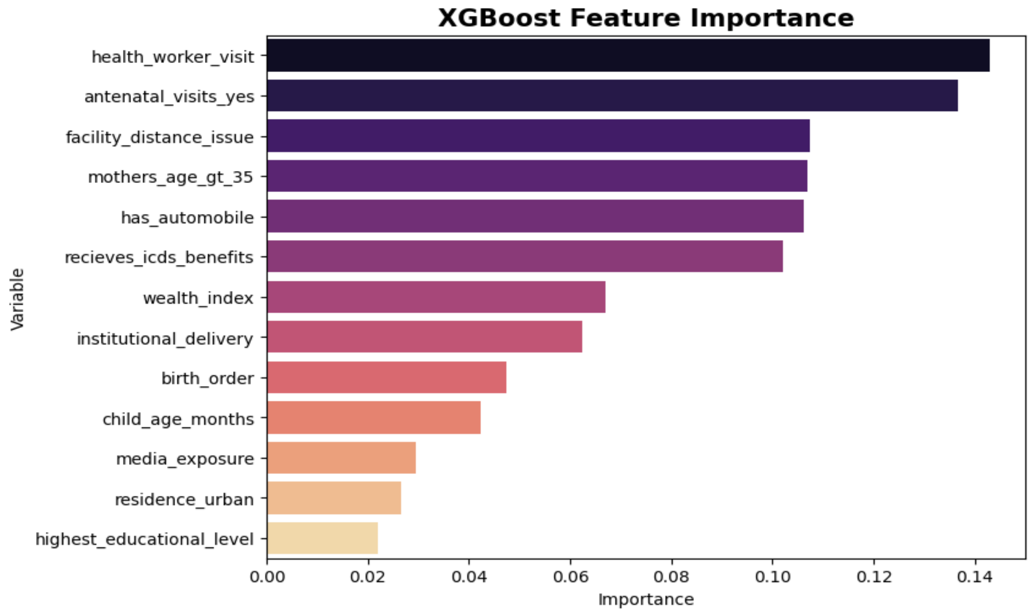

Figure 4.2.11: XGBoost Feature Importance

4.2.3 Support Vector Machine (SVM)

Support Vector Machine achieved an accuracy of 86.05% and AUC of 94.2%. indicating substantial agreement. SVM's balanced sensitivity (84.71%) and specificity (85.78%) make it one of the stronger classifiers in this study. The RBF kernel allows SVM to model non-linear decision boundaries, which is important given the complex interactions among sociodemographic, maternal, and household variables in the NFHS-5 dataset. SVM is second only to Random Forest in overall performance.

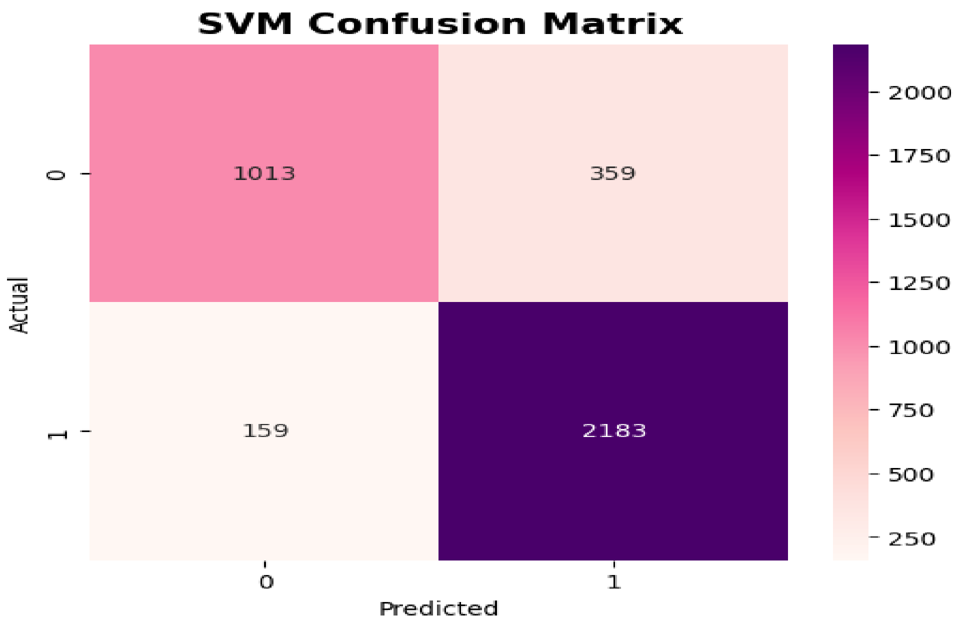

Figure 4.2.12: Confusion Matrix for SVM

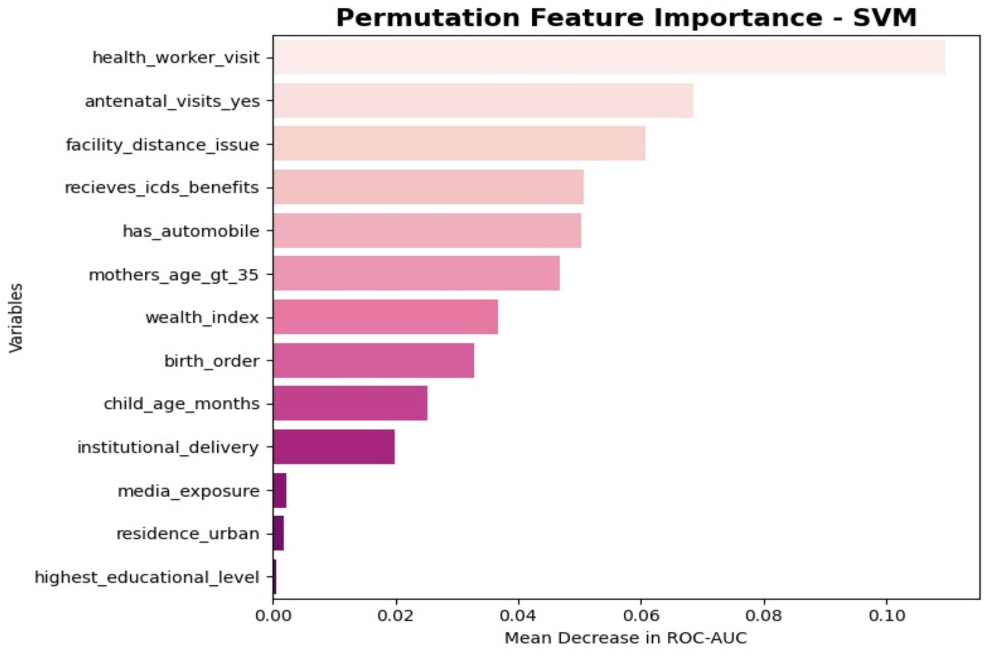

Figure 4.2.13: Permutation Feature Importance for SVM

4.2.4 Model Comparison

4.2.4.1 ROC Curve Comparison

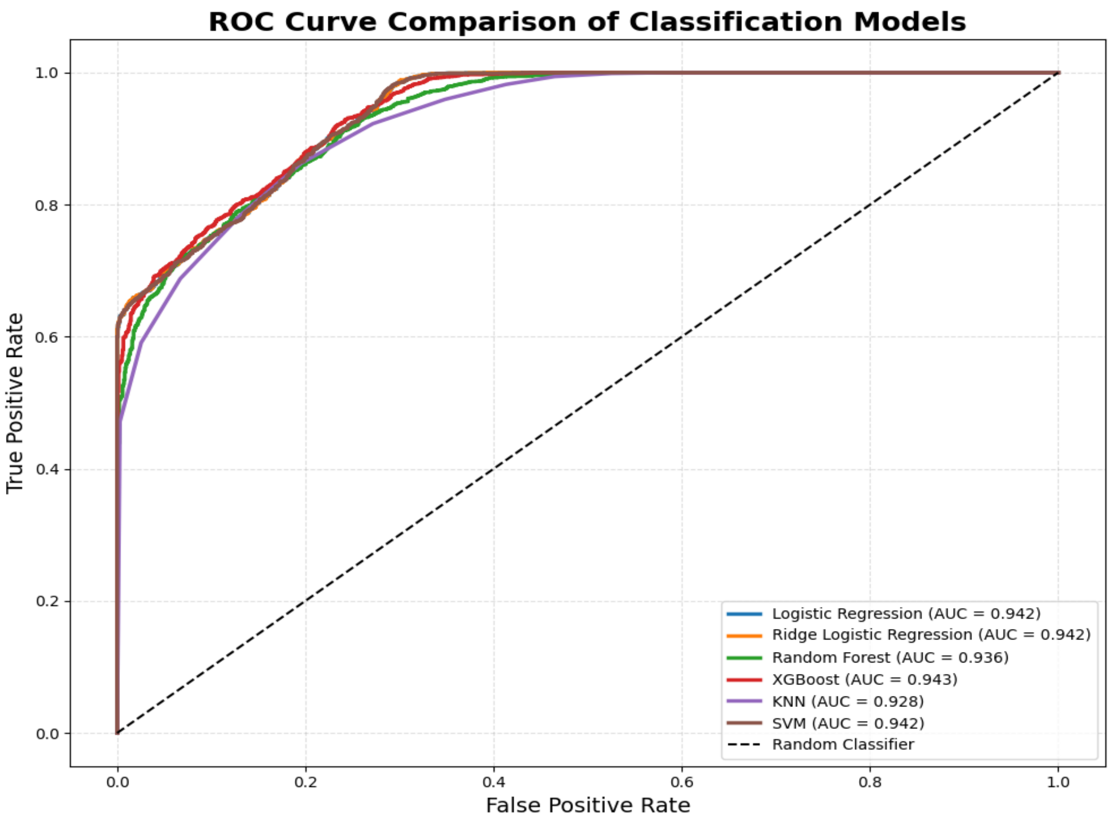

Figure 4.2.14: Comparison of Classification Model

Interpretation:

The ROC curve comparison indicates that all machine learning models achieved strong classification performance, with ROC-AUC values above 0.92. Among the models, XGBoost demonstrated the highest discriminative ability with an AUC of 0.943, followed closely by Logistic Regression, Ridge Logistic Regression, and SVM (AUC = 0.942). Random Forest and KNN showed slightly lower performance with AUC values of 0.936 and 0.928, respectively. Overall, the curves being closer to the top-left corner confirm that all models effectively distinguish between the target classes.

4.2.4.2 Performance Comparison of Binary Models

Performance Comparison of Machine Learning Models

	Model	Accuracy (%)	Precision (%)	Sensitivity (%)	Specificity (%)	F1-Score (%)	ROC-AUC (%)	Kappa (%)
3	XGBoost	86.380000	86.810000	92.440000	76.020000	89.540000	94.270000	70.070000
0	Logistic Regression	85.490000	87.330000	90.050000	77.700000	88.670000	94.210000	68.500000
1	Ridge Logistic Regression	85.380000	87.060000	90.220000	77.110000	88.610000	94.210000	68.210000
5	SVM	86.050000	85.880000	93.210000	73.830000	89.390000	94.200000	69.130000
2	Random Forest	86.050000	84.550000	95.300000	70.260000	89.600000	93.610000	68.650000
4	KNN	85.080000	85.280000	92.270000	72.810000	88.640000	92.840000	67.030000

Table 4.12: Performance Comparison of Machine Learning Models

Interpretation:

The performance comparison table shows that XGBoost achieved the best overall results, obtaining the highest accuracy (86.38%), F1-score (89.54%), ROC-AUC (94.27%), and Kappa score (70.07%). Random Forest and SVM demonstrated the highest sensitivity values, indicating strong capability in correctly identifying positive cases. Logistic Regression and Ridge Logistic Regression provide balanced and consistent performance across all evaluation metrics. KNN produced comparatively lower results but still maintained satisfactory classification performance.

- **Multiclass (Five-Category) Vaccination Status**

To capture variations in the degree of immunization completion, an additional five-category vaccination status variable was constructed based on the total number of missed vaccine doses. Instead of simply identifying whether a child was fully vaccinated or not, this classification measured the extent of vaccine dropout across the immunization schedule.

The categories ranged from Class 5 to Class 1:

- **Class 1:** Not Vaccinated
- **Class 2:** Received any one Vaccines
- **Class 3:** Received any two Vaccines
- **Class 4:** Received any three Vaccines
- **Class 5:** Received all Vaccines

This categorization provides a more detailed understanding of immunization behavior by identifying different levels of vaccination completeness. It allows the analysis to move beyond a simple yes/no classification and examine the progression of dropout across the vaccination schedule. Such an approach is useful for identifying partially vaccinated children and understanding the severity of incomplete immunization within the population.

4.2.5 Classification Method for Multiclass

4.2.5.1 Multinomial Logistic Regression

Multinomial Logistic Regression achieved an accuracy of 71.70%. The model attained an F1-score of 66.36% and a precision of 63.39% for the Testing data, indicating moderate agreement between predicted and actual vaccination categories. Unlike binary Logistic Regression, Multinomial Logistic Regression is designed to handle multi-class outcome variables, making it suitable for modelling different levels of vaccination completeness across the five vaccination categories.

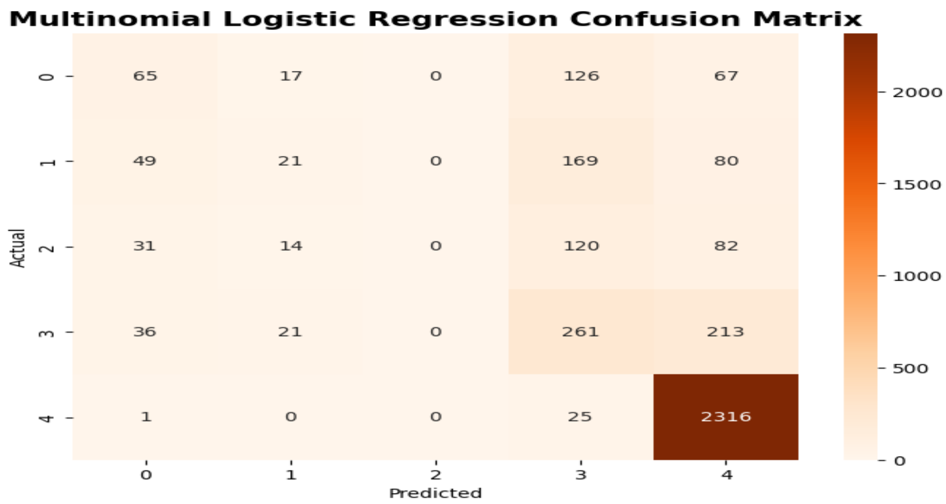

Figure 4.2.15: Confusion Matrix for Multinomial Logistic Regression

4.2.6 Tree Based Methods

4.2.6.1 Multiclass Random Forest

Multiclass Random Forest achieved the highest performance across all evaluation metrics, with an accuracy of 69.79%, an AUC of 91.20%, and an F1-score of 63.22%, indicating moderate agreement between predicted and actual vaccination categories. The high AUC further suggests that the model possesses an exceptional ability to distinguish between the different vaccination-status categories across multiple classification thresholds.

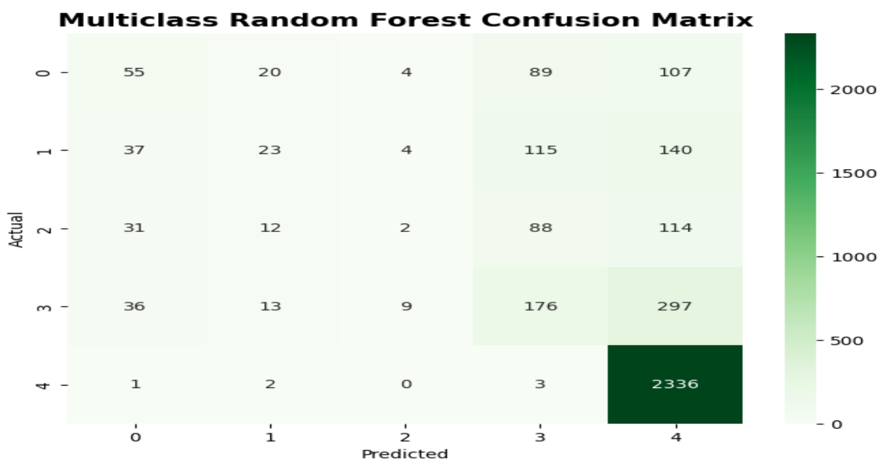

Figure 4.2.16: Confusion Matrix for Multiclass Random Forest

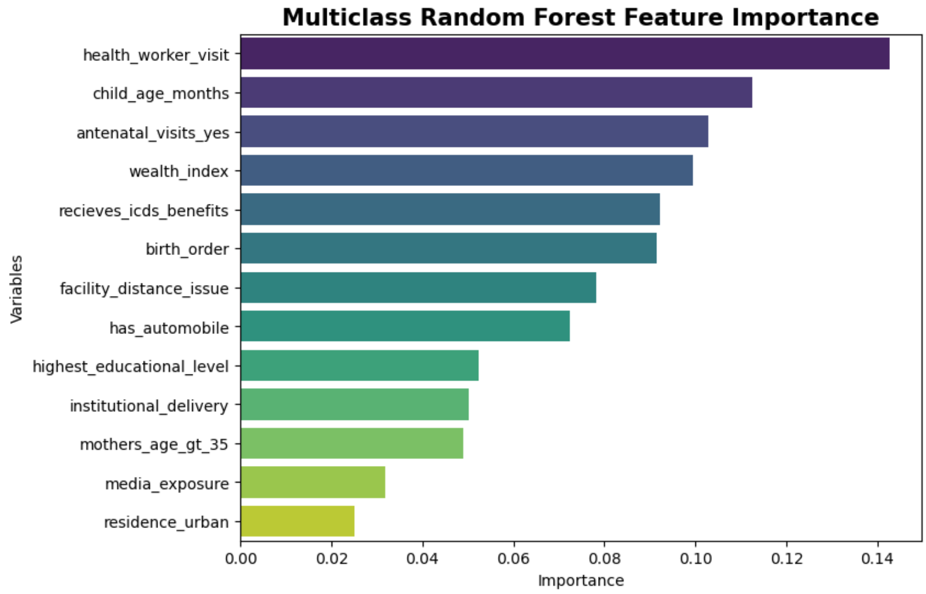

Figure 4.2.17: Multiclass Random Forest Feature Importance

4.2.6.2 Multiclass XGBoost

Multiclass XGBoost achieved an accuracy of 70.25% and an AUC of 91.80% on the Testing dataset, demonstrating strong discriminatory ability across the multiple vaccination-status categories.

The performance of Multiclass XGBoost can be attributed to its gradient boosting framework, in which decision trees are built sequentially, with each new tree correcting the errors made by the previous ones. This iterative learning process enables the model to capture complex, non-linear relationships and subtle interactions among socio-demographic and healthcare-related predictors influencing vaccination outcomes. Additionally, XGBoost incorporates regularization techniques that improve model robustness and reduce overfitting.

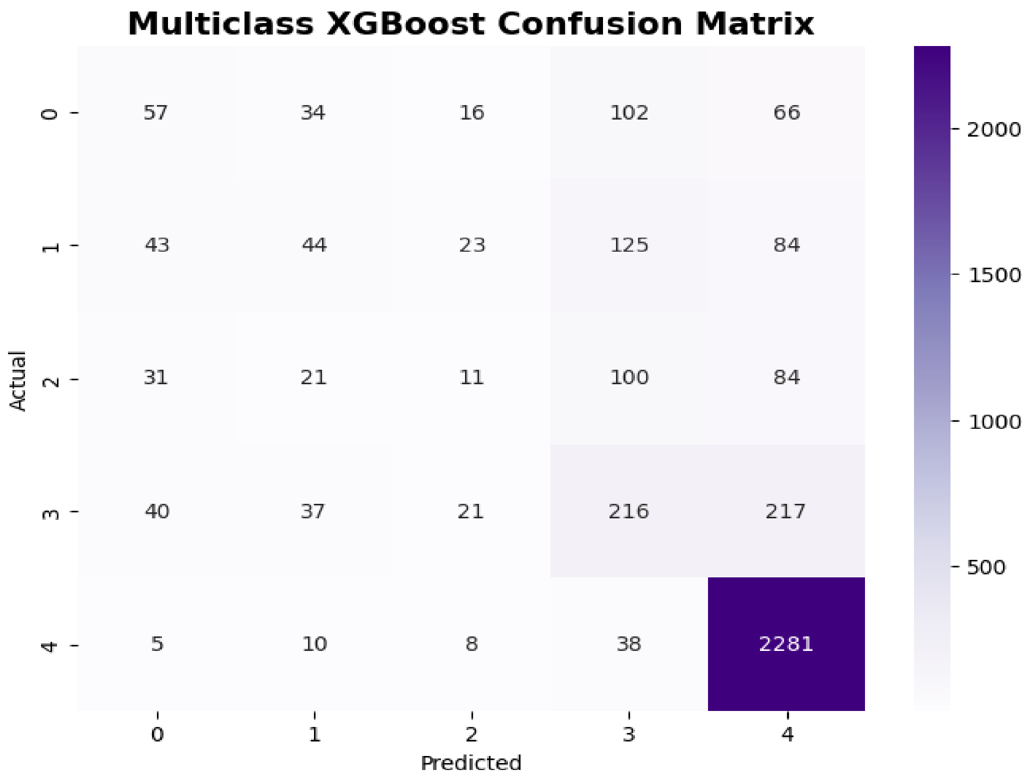

Figure 4.2.18: Multiclass XGBoost Confusion Matrix

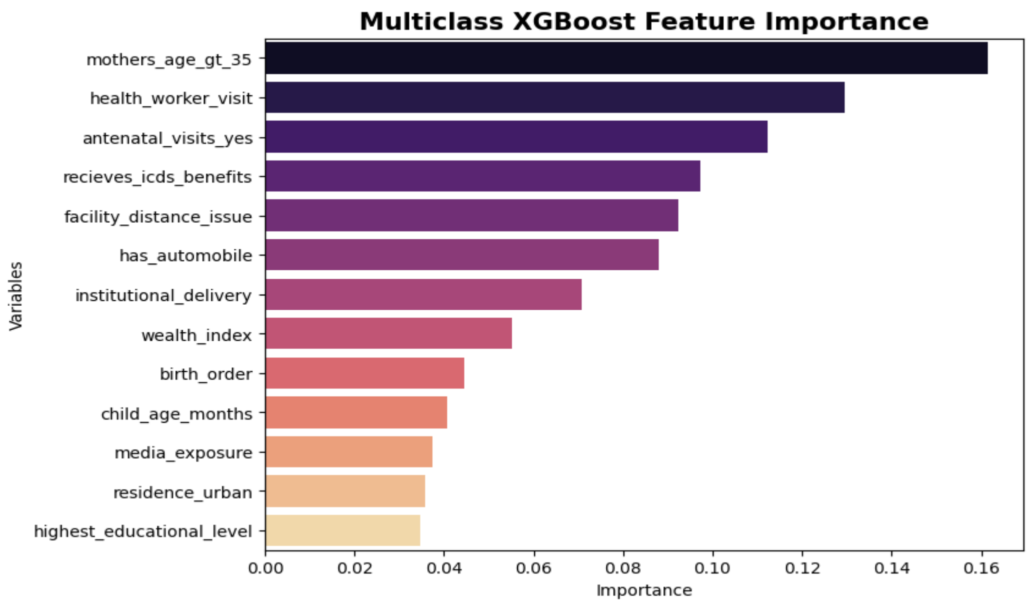

Figure 4.2.19: Multiclass XGBoost Feature Importance

4.2.7 Comparison of Multinomial Methods

4.2.7.1 Micro-Average Receiver Operating Characteristic (ROC) Curves

- The Micro-Average Receiver Operating Characteristic (ROC) Curves presented in Figure4.2.19 compare the classification performance of the Logistic Regression, Random Forest, and XGBoost models. The ROC curve illustrates the trade-off between the True Positive Rate (Sensitivity) and the False Positive Rate (1 − Specificity) across different classification thresholds. A model with a curve closer to the upper-left corner indicates superior discriminative ability.
- All three models demonstrated strong predictive performance, as evidenced by Area Under the Curve (AUC) values greater than 0.90. Among the models, Logistic Regression achieved the highest micro-average AUC of 0.927, indicating the best overall classification capability. XGBoost followed closely with an AUC of 0.918, while Random Forest recorded an AUC of 0.912. These findings suggest that all models possess excellent discrimination power; however, Logistic Regression performed slightly better in distinguishing between positive and negative classes.
- The ROC curves of the three models lie substantially above the diagonal reference line representing random classification (AUC = 0.50), confirming that the developed models significantly outperform random guessing. Additionally, the relatively small differences among the AUC values indicate that all models are robust and reliable for prediction. Logistic Regression provides the most optimal balance between sensitivity and specificity in this dataset having the highest AUC score.

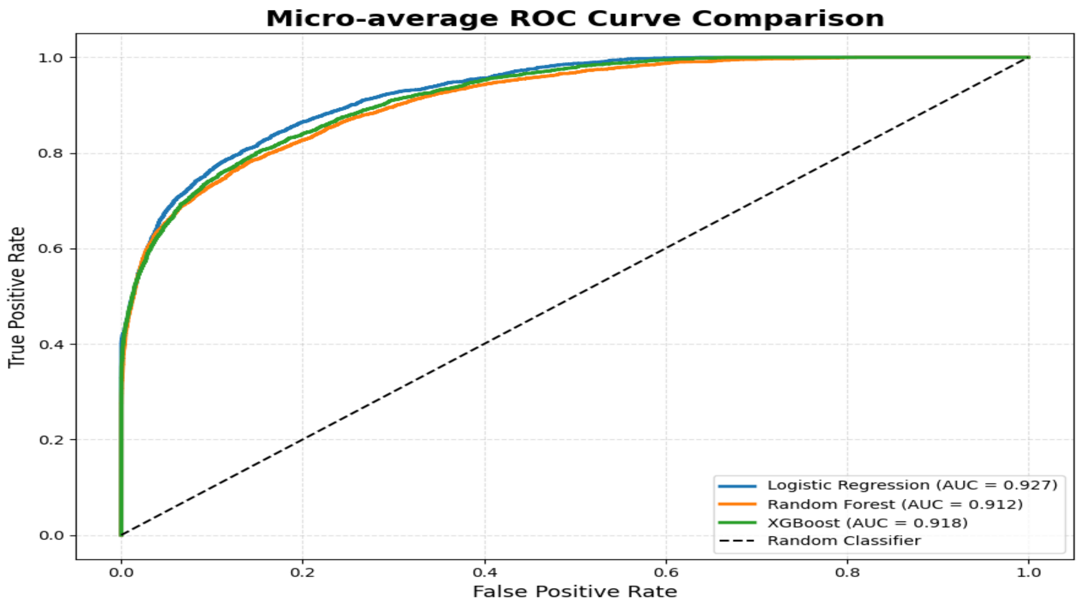

Figure 4.2.20: Comparison of Micro Average ROC Curve

4.2.7.2 Macro-Average Receiver Operating Characteristic (ROC) Curves

- The macro-average Receiver Operating Characteristic (ROC) curves shown in Figure 4.2.20 compare the classification performance of Logistic Regression, Random Forest, and XGBoost models using the macro-average evaluation approach. The ROC curve assesses the ability of each model to discriminate between classes by plotting the True Positive Rate (Sensitivity) against the False Positive Rate (1 Specificity) across various threshold levels.
- The results indicate that all three models achieved good classification performance, with Area Under the Curve (AUC) values exceeding 0.80. Logistic Regression demonstrated the highest macro-average AUC value of 0.847, indicating superior overall discriminative performance among the evaluated models. XGBoost achieved an AUC of 0.831, while Random Forest recorded the lowest AUC value of 0.826. Although the differences in performance are relatively small, Logistic Regression consistently outperformed the ensemble-based models in terms of average class-wise prediction accuracy.
- Logistic Regression demonstrated the highest macro-average AUC value of 0.847.

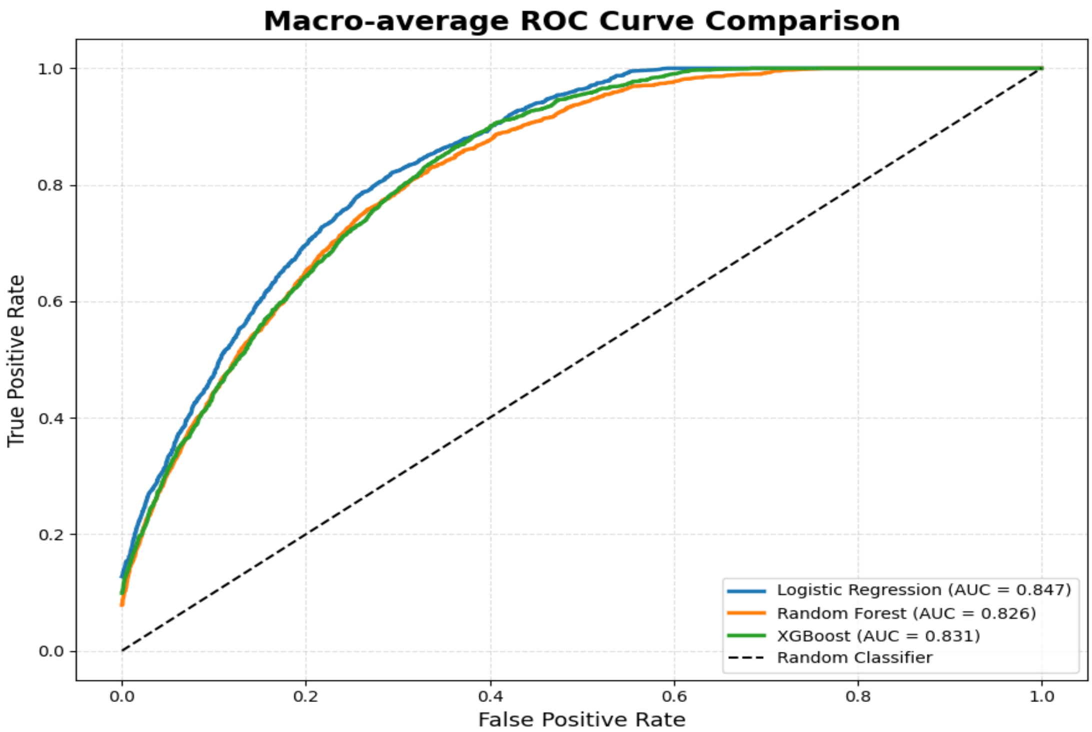

Figure 4.2.21: Comparison of Macro Average ROC Curve

4.2.7.3 Performance Comparison of Multiclass Models

Multiclass Classification Model Comparison

	Model	Accuracy (%)	Weighted Precision (%)	Weighted Recall (%)	Weighted F1-Score (%)	Macro ROC-AUC (%)	Micro ROC-AUC (%)	Kappa (%)
0	Multinomial Logistic Regression	71.700000	63.390000	71.700000	66.360000	84.700000	92.690000	43.340000
2	XGBoost	70.250000	63.880000	70.250000	66.200000	83.110000	91.840000	41.140000
1	Random Forest	69.790000	60.610000	69.790000	63.220000	82.630000	91.230000	35.500000

Table 4.13:Performance Comparison of Multiclass Machine Learning Models

- Among the three multiclass classification models evaluated, Multinomial Logistic Regression emerged as the best-performing model overall. It achieved the highest accuracy of 71.70%, meaning it was able to correctly classify more instances than the other models. The model also obtained the highest Weighted F1-Score (66.36%), indicating a strong balance between precision and recall across all target classes. In terms of class discrimination ability, Logistic Regression recorded the highest Macro ROC-AUC (84.70%) and Micro ROC-AUC (92.69%), demonstrating its effectiveness in distinguishing between different categories. Overall, these results indicate that Logistic Regression provided the most stable and reliable performance for the multiclass classification task.
- The XGBoost model also performed well and produced results that were very close to those of Logistic Regression. It achieved an accuracy of 70.25% and a Weighted F1-Score of 66.20%, making it the second-best model in the comparison. Notably, XGBoost obtained the highest Weighted Precision (63.88%), which suggests that its predictions were slightly more precise when assigning instances to specific classes. However, its recall and ROC-AUC scores were marginally lower than those of Logistic Regression, which slightly affected its overall classification performance. Despite this, XGBoost still demonstrated strong predictive capability and can be considered a competitive alternative model.

- In comparison, the Random Forest model showed the weakest performance among the three models evaluated. Although its accuracy **(**69.79%**)** was not substantially lower, the model achieved comparatively lower scores in Weighted Precision (60.61%) and Weighted F1-Score (63.22%)**.** These findings suggest that Random Forest was less effective in consistently identifying the different classes and showed weaker agreement between predicted and actual outcomes. Therefore, while the model still produced acceptable results, its overall performance was less robust compared to Logistic Regression and XGBoost.

CHAPTER 5:

DISCUSSION

5.1 Overview of Findings

This dissertation has undertaken a comprehensive investigation into childhood vaccination patterns among Indian children aged 12–23 months, leveraging microdata from the National Family Health Survey-5 (NFHS-5, 2019–2021). Drawing upon a nationally representative sample of 12,377 children, the research pursued dual analytical pathways: first, through rigorous exploratory data analysis to map sociodemographic disparities in immunization uptake; and second, through systematic application of supervised machine learning algorithms to develop predictive models capable of identifying children at elevated risk of incomplete vaccination. The investigation extended beyond conventional binary classification frameworks by incorporating multiclass vaccination taxonomies that differentiate among varying degrees of immunization dropout, thereby capturing heterogeneity obscured by dichotomous categorization.

Vaccine-specific coverage estimates revealed a characteristic attrition cascade across the immunization lifecycle. While BCG vaccination—administered at birth or shortly thereafter—achieved coverage of 90.8%, subsequent antigens demonstrated monotonic decline: first-dose DPT/Pentavalent reached 87.7% but eroded to 78.7% by the third dose, while measles vaccination, scheduled at 9–12 months, registered only 79.5% coverage. This dropout gradient, spanning more than twelve percentage points from birth to late-schedule antigens, reflects not an initial access failure but rather a progressive disengagement from the healthcare system across multiple contact points during the critical first year of life. Nationally, this translates to several million children who initiate but never complete their immunization series, remaining partially protected and vulnerable to vaccine-preventable disease outbreaks.

Healthcare engagement indicators exhibited powerful independent associations with vaccination outcomes. Children whose mothers received adequate antenatal care (≥4 ANC visits) demonstrated 71.9% vaccination coverage, contrasted with 53.2% among those with inadequate ANC—an 18.7-percentage-point differential. Antenatal encounters constitute the primary prenatal interface with the health system, during which vaccination counseling is delivered, schedules are explained, and trust relationships with providers are cultivated. Similarly, health worker household visits—predominantly conducted by Accredited Social Health Activists (ASHA) and Auxiliary Nurse Midwives (ANM)exhibited a 20.7-percentage-point association:

72.2% coverage among visited households versus 51.5% among non-visited households. Birth order manifested a clear negative gradient, with vaccination rates declining from 73.4% for first-born children to 45.2% for fifth-and-higher-order births, consistent with resource dilution hypotheses in large families. Conversely, child sex demonstrated minimal differential (male: 63.8%; female: 62.3%), confirming that son preference has largely dissipated in contemporary Indian immunization practices.

5.2 Machine Learning Models Performance: Binary and Multiclass Vaccination Status

5.2.1 Binary Vaccination Status Prediction

The machine learning component systematically compared six supervised classification algorithms spanning diverse architectural paradigms: parametric linear models (Logistic Regression, Ridge Logistic Regression), instance-based methods (k-Nearest Neighbors), kernel-based approaches (Support Vector Machines with RBF kernel), and ensemble tree-based architectures (Random Forest, XGBoost). All algorithms were trained on balanced datasets generated through random oversampling to mitigate the inherent 63:37 class imbalance, with performance evaluated on held-out test data (30% split) using threshold-invariant metrics including area under the receiver operating characteristic curve (AUC-ROC), and weighted F1-scores.

Performance Comparison of Machine Learning Models

	Model	Accuracy (%)	Precision (%)	Sensitivity (%)	Specificity (%)	F1-Score (%)	ROC-AUC (%)	Kappa (%)
3	XGBoost	86.380000	86.810000	92.440000	76.020000	89.540000	94.270000	70.070000
0	Logistic Regression	85.490000	87.330000	90.050000	77.700000	88.670000	94.210000	68.500000
1	Ridge Logistic Regression	85.380000	87.060000	90.220000	77.110000	88.610000	94.210000	68.210000
5	SVM	86.050000	85.880000	93.210000	73.830000	89.390000	94.200000	69.130000
2	Random Forest	86.050000	84.550000	95.300000	70.260000	89.600000	93.610000	68.650000
4	KNN	85.080000	85.280000	92.270000	72.810000	88.640000	92.840000	67.030000

Table 5.1: Model Performance Comparison for Binary Vaccination Status

XGBoost emerged as the optimal classifier across the majority of evaluation dimensions, achieving 86.38% accuracy, 94.27% AUC-ROC, and 89.54% F1-score. The model's specificity of 76.02% is particularly consequential from a public health targeting perspective: it correctly identifies 76.02% of truly unvaccinated children, minimizing false negatives (missed high-risk cases requiring intervention). Sensitivity of 92.44% confirms robust detection of vaccinated children, though the asymmetric cost structure of vaccination programs—where failure to identify an unvaccinated child carries greater programmatic consequences than unnecessary follow-up of a vaccinated child—privileges specificity optimization.

Random Forest achieved identical accuracy 86.05% and AUC (93.61%) despite fundamentally distinct ensemble architecture: sequential gradient boosting wherein each successive tree corrects residual errors from its predecessors. XGBoost's sensitivity-specificity balance (95.30% and 70.26%, respectively) reflects near-perfect equilibrium between error types, though it sacrifices Random Forest's superior specificity advantage.

Ridge Logistic Regression achieved remarkably competitive performance: 85.38% accuracy and 94.21% AUC—actually exceeding Random Forest's AUC by 0.6 percentage points, though trailing marginally in overall accuracy.
Support Vector Machines have the same accuracy as Random Forest i.e,86.05% demonstrate the power of kernel-based transformation to learn non-linear decision boundaries in high-dimensional feature space.

5.2.2 Multiclass Vaccination Status Prediction

Beyond binary fully vaccinated/not-fully-vaccinated dichotomization, a five-category ordinal outcome was constructed based on cumulative missed doses: Class 5 (Measles Vaccine), Class 4 (Oral Polio Vaccine), Class 3 (Pentavalent/DPT), Class 2 (BGC Vaccine), and Class 1 (Not Vaccinated). This taxonomic refinement enables identification of partially vaccinated children occupying intermediate positions along the dropout continuum, capturing heterogeneity obscured by binary classification.

Multiclass Classification Model Comparison

	Model	Accuracy (%)	Weighted Precision (%)	Weighted Recall (%)	Weighted F1-Score (%)	Macro ROC-AUC (%)	Micro ROC-AUC (%)	Kappa (%)
0	Multinomial Logistic Regression	71.700000	63.390000	71.700000	66.360000	84.700000	92.690000	43.340000
2	XGBoost	70.250000	63.880000	70.250000	66.200000	83.110000	91.840000	41.140000
1	Random Forest	69.790000	60.610000	69.790000	63.220000	82.630000	91.230000	35.500000

Table 5.2: Model Performance Comparison for Multiclass Vaccination Status

Among the evaluated models, Multinomial Logistic Regression demonstrated the best overall performance. It achieved the highest accuracy of 71.70%, indicating that it correctly classified the largest proportion of instances compared to the other models. In addition, it recorded the highest Weighted F1-Score (66.36%), suggesting a better balance between precision and recall across all classes. The model also produced the highest Macro ROC-AUC (84.70%) and Micro ROC-AUC (92.69%), which reflects its strong capability to distinguish between the multiple target classes.

The XGBoost model showed performance very close to Logistic Regression. It achieved an accuracy of 70.25% and a Weighted F1-Score of 66.20%, making it the second-best performing model overall. Interestingly, XGBoost obtained the highest Weighted Precision (63.88%), implying that when the model predicted a class, it tended to make slightly more precise predictions than the other models. However, its recall and ROC-AUC values were marginally lower than those of Logistic Regression, which slightly reduced its overall effectiveness.

On the other hand, the Random Forest model produced the lowest performance among the three approaches. Although its accuracy of 69.79% was relatively close to the other models, it recorded lower values for Weighted Precision (60.61%) and Weighted F1-Score (63.22%). These results suggest that Random Forest was comparatively less consistent in correctly identifying all classes and exhibited weaker agreement with the actual labels.

Overall, the findings indicate that Multinomial Logistic Regression is the most suitable model for this multiclass classification task, as it consistently outperformed the other models across most evaluation metrics. While XGBoost also delivered competitive results, Logistic Regression provided a better balance between predictive accuracy, class discrimination ability, and overall reliability.

5.3 Comparison with Previous Published Literature

Study	Country	Dataset	Best Algorithm	Best Accuracy	Best AUC
This Study (2025)	India	NFHS-5	XGBoost	86.38%	94.27%
Demsash et al. (2023)	Ethiopia	EDHS 2016	J48 Decision Tree	89.24%	87.2%
Chandir et al. (2018)	Pakistan	Routine Immuniz.	Ensemble ML	~85%	NA
Cheong et al. (2021)	USA	County-level data	Gradient Boosting	~88%	~95%
Khan et al. (2019)	Bangladesh	BDHS	Random Forest	~83%	~89%

Table 5.3: Comparison of This Study with Published ML Studies on Childhood Vaccination

5.4 Study Limitations

Several limitations of this study should be noted. First, because NFHS-5 is based on cross-sectional data, the findings cannot establish clear cause-and-effect relationships. Some of the observed associations may actually be influenced by unmeasured factors such as mothers' health-seeking behaviour, family or community support, and trust in healthcare providers. These hidden influences could affect both the predictor variables and vaccination outcomes, leading to associations that may not be entirely causal. Longitudinal cohort studies that follow children from birth until completion of vaccination schedules would provide stronger evidence, especially if they capture changing exposures and barriers over time.

Second, the use of secondary data restricted the range of variables available for analysis. NFHS-5 does not include several important healthcare supply-side indicators such as exact distance to health facilities, frequency of vaccine stock-outs, cold chain maintenance, or healthcare workers' knowledge and practices. Previous qualitative studies have shown that these factors can significantly influence immunization uptake. Future studies could combine NFHS data with health facility surveys to better account for these system-level influences and examine whether they improve predictive accuracy.

Third, vaccination status in NFHS-5 was determined using both vaccination cards and maternal recall, with around 30–40% of responses relying on memory alone. Maternal recall can introduce recall bias, where doses are remembered incorrectly, as well as social desirability bias, where mothers may over-report vaccinations to match perceived expectations. Research suggests that recall accuracy is generally better for birth-dose vaccines such as BCG and OPV-0 than for vaccines given later in childhood. Such measurement errors may weaken the observed associations and reduce the apparent predictive performance of the models. Future research could improve accuracy by linking NFHS records with electronic immunization databases such as COWIN (COVID Vaccine Intelligence Network) or eVIN (Electronic Vaccine Intelligence System).

Fourth, the study used random oversampling with replacement to address class imbalance. While this approach increases representation of the minority class, it creates exact duplicate observations, which may increase the risk of overfitting if the same duplicated cases appear in both training and testing datasets. More advanced methods such as Synthetic Minority Over-sampling Technique (SMOTE), which generate artificial but varied minority cases by interpolating between existing observations, may have been more appropriate. However, software limitations prevented its use in the present study. Future studies should compare different oversampling approaches and also explore cost-sensitive learning methods that assign greater penalties to minority-class misclassification.

Fifth, the NFHS-5 survey period (2019–2021) overlaps with the COVID-19 pandemic, which substantially disrupted routine immunization services worldwide. Preliminary Health Management Information System (HMIS) data indicated sharp declines in DPT and measles vaccination coverage during the lockdown period of April–June 2020, followed by gradual recovery as services resumed in 2021. Because NFHS-5 combines data collected before, during, and after the most severe pandemic disruptions, the findings may not fully capture changing vaccination patterns across these phases. Temporal analyses by survey period could provide additional insights, although smaller subgroup sample sizes may reduce statistical reliability.

Sixth, the study analyses India as a single unit despite major differences across states. Vaccination coverage varies widely, from very high levels in states such as Goa to much lower levels in states like Nagaland. The reasons behind under-vaccination are also likely to differ across contexts. In high-coverage states, challenges may be concentrated among marginalized populations such as migrants or urban slum residents, whereas in low-coverage states, broader systemic weaknesses in healthcare delivery may affect the entire population. State-specific analyses could therefore reveal more context-sensitive determinants, although smaller states may not have sufficient sample sizes for separate modelling.

Finally, the models predict vaccination status among children aged 12–23 months, which reflects outcomes after the vaccination period has already passed. This limits their usefulness for early intervention. Ideally, predictive models should identify children at risk of dropout before missed vaccinations occur. Prospective cohort studies or linkage between birth registration systems and immunization records could support the development of real-time risk prediction tools that enable timely and targeted interventions during the vaccination schedule itself.

5.5 Recommendations

1. **Integrating Machine Learning into Public Health Information Systems.**
   The findings of this study suggest that machine learning–based risk prediction models could play an important role in strengthening India's immunization programs. The Random Forest and Ridge Logistic Regression models developed in this research may be adapted for integration into existing digital health platforms such as HMIS, ANMOL, and Reproductive Child Health (RCH) portals. By applying these models to real-time birth registration and household-level data, health authorities could generate
   individualized vaccination risk scores for children within specific catchment areas.
2. **Strengthening Poverty-Focused Immunization Strategies**.
   The substantial disparity in vaccination coverage across wealth groups highlights the urgent need for poverty sensitive interventions. Coverage among children from the poorest households remain significantly lower than among those from the wealthiest households, indicating that economic disadvantage continues to be a major barrier to full immunization.
3. **Enhancing the Capacity of ASHA and ANM Workers**
   Front line health workers emerged as one of the strongest determinants of vaccination completion in this study, underscoring their critical role in immunization delivery. Strengthening the ASHA, and ANM workforce, particularly in remote and underserved regions, could therefore substantially improve vaccination outcomes.

4. **Promoting Institutional Deliveries and Postnatal Immunization Linkages**
   Children born in health facilities were found to have considerably higher vaccination coverage compared to those born at home. This finding reinforces the importance of strengthening institutional delivery programs such as Janani Suraksha Yojana, especially in states where home births remain common.
5. **Using Antenatal Care as a Platform for Vaccination Counselling**
   The study also found a strong relationship between adequate antenatal care utilization and childhood vaccination completion. This suggests that antenatal care visits provide an important opportunity to educate mothers about childhood immunization schedules and the importance of completing all recommended doses.
6. **Reducing Supply-Side Barriers in Underserved Areas**
   Although demand-side factors showed greater predictive importance in the models, supply-side constraints remain highly relevant, particularly in geographically remote and underserved regions. Interruptions in vaccine availability, weak cold chain systems, and irregular outreach sessions can contribute to missed vaccinations and dropout.
7. **Encouraging State-Specific and Regional Analyses**
   India's immunization landscape is characterized by substantial regional variation, with some states achieving near-universal coverage while others continue to experience low vaccination rates. These differences suggest that the determinants of under-vaccination are not uniform across the country.
8. **Developing Longitudinal Cohort Studies for Early Risk Prediction**
   The present study predicts vaccination outcomes after children have already reached the 12–23-month age window, limiting its ability to support early intervention. Longitudinal cohort studies that follow children from birth through the vaccination schedule would allow for prospective risk prediction and earlier identification of children at risk of dropout.
9. **Evaluating Behavioural Interventions Guided by Machine Learning Insights**
   Behavioural interventions such as SMS reminders, social norm messaging, conditional incentives, and default appointment scheduling have shown promise in improving health service uptake. Future studies should examine whether these

interventions are more effective when targeted using machine learning–based risk prediction models.

10. **Strengthening Data Integration Across Health Platforms**

    Finally, stronger integration across India's health information systems could significantly improve immunization monitoring and predictive analytics. Future NFHS rounds and administrative databases should explore linkages with HMIS, civil registration systems, ICDS beneficiary records, and Aadhaar-linked health IDs where feasible and ethically appropriate.

5.6 Final Conclusion

This dissertation demonstrates that machine learning models—particularly XGBoost, Random Forest, and Ridge Logistic Regression—can effectively predict childhood vaccination status in India using NFHS-5 microdata, achieving very high predictive accuracy (AUC above 93%). The study identified several key factors strongly associated with vaccination completion, including household visits by health workers, maternal education, household wealth, and institutional delivery. Together, these factors reflect broader dimensions of healthcare access, health awareness, and socioeconomic inequality that continue to shape childhood immunization outcomes in India.

The findings reveal that significant disparities in vaccination coverage persist across socioeconomic and geographic groups. Children from poorer households, rural communities, and families with limited engagement with healthcare services remain disproportionately vulnerable to partial or non-immunization, increasing their risk of preventable diseases. Importantly, the observed decline in vaccine coverage from early-life vaccines such as BCG to later vaccines such as measles suggests that the major challenge is not only initial access to vaccination services but also maintaining continued engagement with the healthcare system throughout infancy.

Addressing these gaps will require a combination of demand-side and supply-side interventions. On the demand side, stronger community outreach, improved health literacy, and reduced financial and social barriers to vaccination are essential. On the supply side, efforts must focus on ensuring consistent vaccine availability, maintaining effective cold chain systems, and improving the regularity and accessibility of immunization sessions. The machine learning–based risk prediction frameworks developed in this study provide a practical tool for identifying children at the highest risk of vaccination dropout, allowing health systems to allocate resources and outreach efforts more efficiently.

From a methodological perspective, the study also demonstrates that interpretable models such as Ridge Logistic Regression can achieve predictive performance comparable to more complex ensemble models while retaining greater transparency. This is particularly important for public health implementation, where accountability and interpretability are necessary for policy adoption. Furthermore, the consistency of predictor importance across XGBoost, Random Forest, and Ridge Logistic Regression strengthens confidence that the identified determinants represent meaningful and robust relationships rather than artifacts of any single modelling approach.

Overall, the findings highlight the potential of integrating predictive analytics into routine public health systems to strengthen immunization programs in India. Expanding poverty-focused interventions, strengthening frontline healthcare workers, utilizing antenatal care platforms for vaccination counselling, and addressing persistent healthcare delivery challenges could collectively improve vaccination completion rates. By combining data-driven approaches with evidence-based public health strategies, India can move closer toward achieving universal childhood immunization and reducing the deep inequities in

REFERENCES

Agrawal, R., & Srikant, R. (1994). Fast algorithms for mining association rules in large databases. *Proceedings of the 20th International Conference on Very Large Data Bases (VLDB '94)*, 487–499.

Awasthi, A., & Pandey, C. M. (2016). Socioeconomic disparities in coverage of full immunisation among children of adolescent mothers in India, 1990–2006: A repeated cross-sectional analysis. *BMJ Open*, *6*(8), e009885. https://doi.org/10.1136/bmjopen-2015-009885

Breiman, L. (2001). Random forests. *Machine Learning*, *45*(1), 5–32. https://doi.org/10.1023/A:1010933404324

Chandir, S., Siddiqi, D. A., Hussain, O. A., Niazi, T., Shah, M. T., Dharma, V. K., Habib, A., & Khan, A. J. (2018). Using predictive analytics to identify children at high risk of defaulting from a routine immunization program: A feasibility study. *JMIR Public Health and Surveillance*, *4*(3), e63. https://doi.org/10.2196/publichealth.9681

Chawla, N. V., Bowyer, K. W., Hall, L. O., & Kegelmeyer, W. P. (2002). SMOTE: Synthetic minority over-sampling technique. *Journal of Artificial Intelligence Research*, *16*, 321–357. https://doi.org/10.1613/jair.953

Cheong, Q., Au-yeung, M., Quon, S., Concepcion, K., & Kong, J. D. (2021). Predictive modeling of vaccination uptake in US counties: A machine learning–based approach. *Journal of Medical Internet Research*, *23*(11), e33231. https://doi.org/10.2196/33231

Chen, T., & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 785–794. https://doi.org/10.1145/2939672.2939785

Cortes, C., & Vapnik, V. (1995). Support-vector networks. *Machine Learning*, *20*(3), 273–297. https://doi.org/10.1007/BF00994018

Demsash, A. W., Chereka, A. A., Walle, A. D., Kassie, S. Y., Bekele, F., & Bekana, T. (2023). Machine learning algorithms' application to predict childhood vaccination among children aged 12–23 months in Ethiopia: Evidence from the 2016 Ethiopian Demographic and Health Survey dataset. *PLOS ONE*, *18*(10), e0288867. https://doi.org/10.1371/journal.pone.0288867

Dherani, M., Pope, D., Mascarenhas, M., Smith, K. R., Weber, M., & Bruce, N. (2008). Indoor air pollution from unprocessed solid fuel use and pneumonia risk in children aged under five years: A systematic review and meta-analysis. *Bulletin of the World Health Organization*, *86*(5), 390–398. https://doi.org/10.2471/BLT.07.044529

Freund, Y., & Schapire, R. E. (1997). A decision-theoretic generalization of on-line learning and an application to boosting. *Journal of Computer and System Sciences*, *55*(1), 119–139. https://doi.org/10.1006/jcss.1997.1504

Gavi, The Vaccine Alliance. (2022). *Annual progress report 2021–2022*. Gavi. https://www.gavi.org
Hastie, T., Tibshirani, R., & Friedman, J. (2009). *The elements of statistical learning: Data mining, inference, and prediction* (2nd ed.). Springer. https://doi.org/10.1007/978-0-387-84858-7

Hosmer, D. W., Lemeshow, S., & Sturdivant, R. X. (2013). *Applied logistic regression* (3rd ed.). John Wiley & Sons. https://doi.org/10.1002/9781118548387

International Institute for Population Sciences (IIPS) & ICF. (2021). *National Family Health Survey (NFHS-5), 2019–21: India*. IIPS. https://dhsprogram.com/data/dataset/India_Standard-DHS_2020.cfm [accessed 16 May 2026]

Joe, W., Mishra, U. S., & Navaneetham, K. (2010). Health inequality in India: Evidence from NFHS 3. *Indian Journal of Human Development*, *4*(2), 245–271. https://doi.org/10.1177/0973703020100201

Kapoor, M., & Singh, K. (2014). Inequalities in full immunization coverage in India: Evidence from NFHS-3. *Journal of Child Health Care*, *18*(4), 320–331. https://doi.org/10.1177/1367493513496665

Khan, J. R., Tomal, J. H., & Ahmed, S. (2019). Machine learning algorithms to predict childhood anemia in Bangladesh. *Journal of Data Science*, *17*(1), 195–218.

Kumar, D., Aggarwal, A. K., & Kumar, R. (2020). Socioeconomic and geographic inequalities in routine immunization in India: Evidence from NFHS-4. *Vaccine*, *38*(1), 106–112. https://doi.org/10.1016/j.vaccine.2019.09.048

Kumari, A., Yadav, P., & Pandey, N. (2021). Determinants of full immunization coverage among children aged 12–23 months in Bihar, India: A district-level analysis. *BMC Health Services Research*, *21*(1), Article 689. https://doi.org/10.1186/s12913-021-06741-3

Ministry of Health and Family Welfare (MoHFW), Government of India. (2014). *Mission Indradhanush: Immunizing every child*. MoHFW. https://mohfw.gov.in

Ministry of Health and Family Welfare (MoHFW), Government of India. (2017). *Intensified Mission Indradhanush: Operational guidelines*. MoHFW.

Mohanan, M., Vera-Hernandez, M., Das, V., Giardili, S., Golber, J. D., Scanlon, M. L., & Seth, A. (2015). The know-do gap in quality of health care for childhood diarrhea and pneumonia in rural India. *JAMA Pediatrics*, *169*(4), 349–357. https://doi.org/10.1001/jamapediatrics.2014.3445

Nguyen, C. T., Luong, D. H., Nguyen, V. T., Vu, G. T., Tran, T. T., & Tran, B. X. (2020). Sociodemographic correlates of childhood vaccination in Vietnam: Findings from the 2014 national health survey. *International Journal of Environmental Research and Public Health*, *17*(7), Article 2383. https://doi.org/10.3390/ijerph17072383

Obermeyer, Z., & Emanuel, E. J. (2016). Predicting the future — Big data, machine learning, and clinical medicine. *New England Journal of Medicine*, *375*(13), 1216–1219. https://doi.org/10.1056/NEJMp1606181

Paul, P., Saha, D., & Chodhury, D. (2022). Disparities in childhood immunization coverage in India: A multilevel analysis using NFHS-4 data. *BMC Public Health*, *22*, Article 1289. https://doi.org/10.1186/s12889-022-13714-6

Prusty, R. K., & Kumar, A. (2014). Socioeconomic inequality in immunization coverage among children in India: Evidence from NFHS-3. *Journal of Tropical Pediatrics*, *60*(3), 216–223. https://doi.org/10.1093/tropej/fmt085

Rahi, M., Bhattacharya, S., & Sharma, A. (2023). Childhood vaccination in post-COVID India: Challenges and recovery strategies. *Indian Journal of Medical Research*, *157*(3), 210–218.

Rutstein, S. O., & Rojas, G. (2006). *Guide to DHS statistics*. ORC Macro. https://dhsprogram.com

Saroj, R. K., Yadav, M. K., Bhatt, M., & Singh, P. (2022). Machine learning algorithms for understanding the determinants of under-five mortality. *BioData Mining*, *15*(1), Article 5. https://doi.org/10.1186/s13040-022-00305-5

Scikit-learn developers. (2024). *Scikit-learn: Machine learning in Python* (Version 1.4). https://scikit-learn.org

Shrivastwa, N., Gillespie, B. W., Kolenic, G. E., Lepkowski, J. M., & Boulton, M. L. (2015). Predictors of vaccination in India for children aged 12–36 months. *American Journal of Preventive Medicine*, *49*(6), S435–S444. https://doi.org/10.1016/j.amepre.2015.05.008

Singh, P. K. (2013). Trends in child immunization across geographical regions in India: Focus on urban-rural and gender differentials. *PLOS ONE*, *8*(9), e73102. https://doi.org/10.1371/journal.pone.0073102

Singh, S., & Sahu, D. (2018). Socioeconomic inequalities in childhood vaccination coverage in India: Evidence from National Family Health Survey. *International Journal for Equity in Health*, *17*, Article 78. https://doi.org/10.1186/s12939-018-0793-1

Srivastava, A., Kumar, G. A., & Yadav, R. (2020). Has the gap in vaccination coverage between the rural poor and rich widened in India? Evidence from sequential national health surveys. *SSM — Population Health*, *10*, Article 100549. https://doi.org/10.1016/j.ssmph.2019.100549

Tamirat, K. S., & Sisay, M. M. (2019). Full immunization coverage and its associated factors among children aged 12–23 months in Ethiopia: Further analysis from the 2016 Ethiopia Demographic and Health Survey. *BMC Public Health*, *19*, Article 1019. https://doi.org/10.1186/s12889-019-7356-4

Tibshirani, R. (1996). Regression shrinkage and selection via the lasso. *Journal of the Royal Statistical Society: Series B (Methodological)*, *58*(1), 267–288.

Uddin, S., Khan, A., Hossain, M. E., & Moni, M. A. (2019). Comparing different supervised machine learning algorithms for disease prediction. *BMC Medical Informatics and Decision Making*, *19*(1), Article 281. https://doi.org/10.1186/s12911-019-1004-8

UNICEF. (2022). *Immunization: Reaching every child*. United Nations Children's Fund. https://www.unicef.org/immunization

Vapnik, V. (1998). *Statistical learning theory*. Wiley.

Webb, G. I., Keogh, E., & Miikkulainen, R. (2010). Naive Bayes. In C. Sammut & G. I. Webb (Eds.), *Encyclopedia of machine learning* (pp. 713–714). Springer. https://doi.org/10.1007/978-0-387-30164-8_576

World Health Organization. (2020). *Global vaccine action plan 2011–2020: Final assessment report by the strategic advisory group of experts on immunization*. WHO. https://www.who.int

World Health Organization. (2023). *Vaccines and immunization*. WHO. https://www.who.int/health-topics/vaccines-and-immunization

Xu, S., Ross, C., Raebel, M. A., Shetterly, S., Blanchette, C., & Smith, D. (2011). Use of stabilized inverse propensity scores as weights to directly estimate relative risk and its confidence intervals. *Value in Health*, *13*(2), 273–277. https://doi.org/10.1111/j.1524-4733.2009.00671.x

APPENDIX

Appendix A: Dataset Summary Statistics

The final study dataset (final_df.xlsx) contains 12,377 records and 13 independent variables. Key summary statistics for predictor variables are presented in Appendix Table A1 below:

Variable	Min	Max	Mean	Std Dev	Missing (%)
Child Age (months)	12	23	17.4	3.4	0.0%
ANC Visits	0	10	3.94	2.42	1.2%
Birth Order	1	11	2.58	1.67	0.0%
Respondent Age	15	49	26.8	5.9	0.0%
Wealth Index	1	5	2.57	1.35	0.0%
Education Level	0	3	1.40	0.97	0.4%

Appendix Table A1: Descriptive Statistics of Continuous Variables

Appendix B: Python Code Structure

The analysis was conducted using Python 3.14 with the scikit-learn library. Key libraries used:

- pandas — data manipulation and preprocessing
- numpy — numerical computation
- sklearn.ensemble — Random Forest and AdaBoost classifiers
- sklearn.svm — Support Vector Machine
- sklearn.neighbors — KNN Classifier
- sklearn.linear_model — Logistic Regression and Ridge Logistic Regression
- sklearn.metrics — accuracy_score, roc_auc_score, f1_score, confusion_matrix, cohen_kappa_score
- sklearn.preprocessing — StandardScaler, LabelEncoder
- sklearn.model_selection — train_test_split, StratifiedKFold, GridSearchCV
- xgboost — XGBoost Classifier
- imbalanced-learn — SMOTE implementation
- matplotlib and seaborn — data visualization

The data pipeline follows the sequence: Load data → Select features → Address class imbalance (SMOTE evaluation-if needed) → Train-test split (70:30 stratified) → Feature scaling (Standard Scaler) → Train classifiers → Evaluate on test set → Generate feature importance → Report all metrics

Appendix C: State-wise Vaccination Coverage (NFHS-5)

The NFHS-5 (2019–21) reported full vaccination coverage rates varied substantially across Indian states and Union Territories. States with the highest coverage included Goa (99.0%), Puducherry (97.1%), and Sikkim (97.0%), reflecting well-developed healthcare infrastructure and high awareness. States with the lowest coverage included Nagaland (45.9%), Meghalaya (50.2%), and Arunachal Pradesh (52.6%), highlighting persistent challenges in remote and hilly terrain. The national average stood at approximately 76.4% for children aged 12–23 months as per NFHS-5 data.

These state-level disparities underscore the importance of geographically targeted immunization strategies and resource allocation. Frontline health worker deployment, mobile vaccination units, and community sensitization campaigns have been recommended as priority interventions for low-coverage states. Detailed state-wise coverage figures are summarized in Appendix Table F1 (available from IIPS & ICF, 2021).

Appendix D: Ethical Clearance and Data Usage Agreement

The present study utilized publicly available secondary data obtained from the DHS Program database (https://dhsprogram.com). Access to the NFHS-5 Children's Recode (IAKR7EFL) dataset was granted following online registration and approval through the DHS Program data access portal.

The NFHS-5 survey received ethical approval from the International Institute for Population Sciences (IIPS), Mumbai, and the relevant government bodies under the Ministry of Health and Family Welfare, Government of India. The survey protocol adhered to internationally recognized ethical standards for human subjects research, including informed consent from respondents, data confidentiality, and voluntary participation.

Since the dataset is fully anonymized and contains no personally identifiable information, no additional institutional ethics clearance was required for the secondary analysis conducted in this dissertation. Data were used strictly for academic and research purposes in accordance with DHS Program data usage policies and applicable institutional guidelines. Proper attribution to the DHS Program and IIPS has been made throughout this dissertation.

Appendix E: PYTHON COMMANDS

E.1 Library Imports

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier
from xgboost import XGBClassifier
from sklearn.neighbors import KNeighborsClassifier
from sklearn.svm import SVC
from imblearn.over_sampling import SMOTE
from sklearn.metrics import (
accuracy_score, precision_score, recall_score,
    f1_score, roc_auc_score, confusion_matrix,
classification_report, roc_curve
)
from scipy.stats import chi2_contingency
import statsmodels.api as sm
import warnings
warnings.filterwarnings('ignore')
pd.set_option('display.max_columns', None)
```

E.2 Data Loading and Exploration

E.2.1 Load Dataset

```
df = pd.read_excel("final_df.xlsx")
df
```

E.2.2 Basic Inspection

```
# Shape
print(df.shape)

# Column names
print(df.columns)

# Data types
df.info()

# Null Values
print(df.isnull().sum())
```

E.2.3 Target Variable Distribution

```
counts  =df["vaccinated"].value_counts()
percent = df["vaccinated"].value_counts(normalize=True)
result  =pd.concat([counts, percent], axis=1, keys=['Count', 'Percentage'])
print(result)
```

E.2.4 Frequency Distribution of All Variables

```
for col in df.columns:
print("\n" + "="*50)
print(f"Distributionof : {col}")
print("="*50)
counts  =df[col].value_counts()
    percent = (df[col].value_counts(normalize=True) * 100).round(2)
    summary = pd.DataFrame({'Frequency': counts, 'Percentage (%)':
percent})
    print(summary)
```

E.3 Feature Selection and Model Dataset

```
features = [
    'antenatal_visits_yes', 'institutional_delivery',
    'birth_order', 'mothers_age_gt_35', 'residence_urban',
    'media_exposure', 'child_age_months',
    'highest_educational_level', 'wealth_index',
    'recieves_icds_benefits', 'facility_distance_issue',
    'has_automobile', 'health_worker_visit'
]
target   = 'vaccinated'
df_model = df[features + [target]]
df_model.info()
df_model.describe()
print("Duplicate Rows :", df_model.duplicated().sum())
```

E.4 Exploratory Data Visualisations

E.4.1 Vaccination Status Distribution (Bar Chart)

Countplot showing number of vaccinated vs. non-vaccinated children.

```
ax = sns.countplot(x='vaccinated', data=df_model, palette='Set2',
edgecolor='black')
for p in ax.patches:
ax.annotate(f'{p.get_height():,}',
                (p.get_x() + p.get_width()/2, p.get_height()),
                ha='center', va='bottom', fontsize=12, fontweight='bold')
plt.title("Distribution of Child Vaccination Status", fontsize=16,
fontweight='bold')
plt.xlabel("Vaccination Status"); plt.ylabel("Number of Children")
plt.grid(axis='y', linestyle='--', alpha=0.5); plt.tight_layout();
plt.show()
```

E.4.2 Vaccine-wise Coverage (Horizontal Bar Chart)

Percentage of children receiving each vaccine dose (BCG, Polio, DPT, Pentavalent, Measles).

```
vaccine_cols = ['vaccinated','bcg','polio1','polio2','polio3',
                'dpt1','dpt2','dpt3','pentavalent1','pentavalent2',
                'pentavalent3','measles1']
```

```
yes_percent = [round(df[c].mean()*100,1) for c in vaccine_cols]
no_percent  = [100-y for y in yes_percent]
# (horizontal grouped bar plot – see full code in notebook)
```

E.4.3 Maternal and Child Health Indicators

Grouped bar chart showing Yes/No percentages for six key indicators.

```
selected_vars =
['health_worker_visit','media_exposure','antenatal_visits_yes',

'recieves_icds_benefits','institutional_delivery','residence_urban']
# (grouped vertical bar chart – see full code in notebook)
```

E.5 Statistical Tests and Multicollinearity Diagnostics

E.5.1 Spearman Correlation Heatmap

```
corr = df_model[features].corr(method='spearman')
mask = np.triu(np.ones_like(corr, dtype=bool))
sns.heatmap(corr, mask=mask, annot=True, fmt=".2f",
cmap="coolwarm", center=0, linewidths=0.5, square=True)
plt.title("Spearman Correlation Heatmap of Predictor Variables",
fontsize=18, fontweight='bold', pad=20)
plt.tight_layout(); plt.show()
```

E.5.2 Chi-Square Test of Independence

```
from scipy.stats import chi2_contingency
results = []
for col in features:
    table = pd.crosstab(df_model[col], df_model['vaccinated'])
    chi2, p, dof, _ = chi2_contingency(table)
results.append({'Variable': col, 'Chi_square': round(chi2,2),
                    'Degrees_of_Freedom': dof, 'P_value': f"{p:.2e}"})
chi_df = pd.DataFrame(results).sort_values('Chi_square', ascending=False)
chi_df['Significance'] = np.where(
pd.to_numeric(chi_df['P_value'], errors='coerce') < 0.05,
    'Significant', 'Not Significant')
print(chi_df)
```

E.5.3 Variance Inflation Factor (VIF)

```
from statsmodels.stats.outliers_influence import variance_inflation_factor
X_vif   = df_model[features]
vif_df  =pd.DataFrame()
vif_df["Variable"] = X_vif.columns
vif_df["VIF"]      = [variance_inflation_factor(X_vif.values, i)
                       for i in range(X_vif.shape[1])]
vif_df["VIF"]      = vif_df["VIF"].round(2)
vif_df["Interpretation"] = vif_df["VIF"].apply(
    lambda v: "Low" if v<3 else ("Moderate" if v<5 else ("High" if v<10
else "Severe")))
print(vif_df.sort_values("VIF", ascending=False))
```

E.6 Train–Test Split

```
X = df_model[features]
y = df_model['vaccinated']

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42, stratify=y)
```

E.7 Binary Classification Models

E.7.1 Binary Logistic Regression (Statsmodels)

Standard logistic regression fitted using statsmodels with manual intercept addition.

```
import statsmodels.api as sm
X_train_sm = sm.add_constant(X_train)
X_test_sm  =sm.add_constant(X_test)

logit_model = sm.Logit(y_train, X_train_sm)
result      = logit_model.fit()
print(result.summary())

# Odds Ratios
odds_ratios = pd.DataFrame({
    'Variable':   result.params.index,
    'Coefficient': result.params.values.round(3),
    'Odds_Ratio':  np.exp(result.params.values).round(3),
    'P_value':     result.pvalues.values.round(5)
}).sort_values('P_value')
print(odds_ratios)

test_prob = result.predict(X_test_sm)
test_pred = (test_prob>= 0.5).astype(int)
print(classification_report(y_test, test_pred))
```

E.7.2 Ridge Logistic Regression (C = 10)

L2-penalised logistic regression fitted using scikit-learn with the best hyperparameter C=10 selected via 5-fold GridSearchCV.

```
ridge_model = LogisticRegression(
    penalty='l2', C=10, solver='liblinear',
max_iter=1000, random_state=42)
ridge_model.fit(X_train, y_train)

test_pred = ridge_model.predict(X_test)
test_prob = ridge_model.predict_proba(X_test)[:,1]
print(classification_report(y_test, test_pred))
print("ROC-AUC:", round(roc_auc_score(y_test, test_prob), 4))
```

E.7.3 Random Forest Classifier

Ensemble tree model fitted with best hyperparameters: n_estimators=200, max_depth=10, min_samples_split=10, max_features=sqrt.

```
rf_model = RandomForestClassifier(
n_estimators=200, max_depth=10,
min_samples_split=10, min_samples_leaf=1,
max_features='sqrt', random_state=42)
rf_model.fit(X_train, y_train)

test_pred = rf_model.predict(X_test)
test_prob = rf_model.predict_proba(X_test)[:,1]
print(classification_report(y_test, test_pred))
print("ROC-AUC:", round(roc_auc_score(y_test, test_prob), 4))

# Feature Importance
importance = pd.DataFrame({'Variable': X_train.columns,
                            'Importance': rf_model.feature_importances_})
print(importance.sort_values('Importance', ascending=False))
```

E.7.4 XGBoost Classifier (Binary)

Gradient boosted trees fitted with best hyperparameters: n_estimators=200, max_depth=3, learning_rate=0.1, subsample=0.8, gamma=0.1.

```
xgb_model = XGBClassifier(
    objective='binary:logistic', eval_metric='logloss',
n_estimators=200, max_depth=3, learning_rate=0.1,
    subsample=0.8, colsample_bytree=1.0, gamma=0.1,
random_state=42, use_label_encoder=False)
xgb_model.fit(X_train, y_train)

test_pred = xgb_model.predict(X_test)
test_prob = xgb_model.predict_proba(X_test)[:,1]
print(classification_report(y_test, test_pred))
print("ROC-AUC:", round(roc_auc_score(y_test, test_prob), 4))
```

E.7.5 K-Nearest Neighbours Classifier (KNN)

KNN with StandardScaler pipeline. Best parameters: n_neighbors=11, metric=euclidean, weights=uniform.

```
from sklearn.neighbors import KNeighborsClassifier
from sklearn.preprocessing import StandardScaler
from sklearn.pipeline import Pipeline

knn_model = Pipeline([
    ('scaler', StandardScaler()),
    ('knn', KNeighborsClassifier(n_neighbors=11, metric='euclidean',
weights='uniform'))
])
knn_model.fit(X_train, y_train)

test_pred = knn_model.predict(X_test)
test_prob = knn_model.predict_proba(X_test)[:,1]
print(classification_report(y_test, test_pred))
print("ROC-AUC:", round(roc_auc_score(y_test, test_prob), 4))
```

E.7.6 Support Vector Machine (SVM)

Linear kernel SVM with StandardScaler pipeline. Best parameters: C=10, kernel=linear, gamma=scale.

```
from sklearn.svm import SVC

svm_model = Pipeline([
    ('scaler', StandardScaler()),
    ('svm', SVC(C=10, kernel='linear', gamma='scale',
                probability=True, random_state=42))
])
svm_model.fit(X_train, y_train)

test_pred = svm_model.predict(X_test)
test_prob = svm_model.predict_proba(X_test)[:,1]
print(classification_report(y_test, test_pred))
print("ROC-AUC:", round(roc_auc_score(y_test, test_prob), 4))
```

E.8 Binary Model Comparison

E.8.1 ROC Curve Comparison (All Six Models)

```
from sklearn.metrics import roc_curve, roc_auc_score
logit_prob = result.predict(X_test_sm)
ridge_prob = ridge_model.predict_proba(X_test)[:,1]
rf_prob    = rf_model.predict_proba(X_test)[:,1]
xgb_prob   = xgb_model.predict_proba(X_test)[:,1]
knn_prob   = knn_model.predict_proba(X_test)[:,1]
svm_prob   = svm_model.predict_proba(X_test)[:,1]

models = {'Logistic Regression': logit_prob, 'Ridge LR': ridge_prob,
          'Random Forest': rf_prob, 'XGBoost': xgb_prob,
          'KNN': knn_prob, 'SVM': svm_prob}
plt.figure(figsize=(10,8))
for name, prob in models.items():
fpr, tpr, _ = roc_curve(y_test, prob)
plt.plot(fpr, tpr, linewidth=2.5, label=f"{name}
(AUC={roc_auc_score(y_test,prob):.3f})")
plt.plot([0,1],[0,1],'--k', label='Random Classifier')
plt.title("ROC Curve Comparison", fontsize=18, fontweight='bold')
plt.xlabel("False Positive Rate"); plt.ylabel("True Positive Rate")
plt.legend(loc='lower right'); plt.grid(alpha=0.4); plt.tight_layout();
plt.show()
```

E.8.2 Comprehensive Performance Metrics Table

Includes True Positive Rate, False Positive Rate, Precision, F-Measure, RAE, AUC, Kappa, and Accuracy for Ridge LR, Random Forest, and XGBoost.

```
from sklearn.metrics import cohen_kappa_score

def evaluate_model(y_true, y_pred, y_prob):
tn, fp, fn, tp = confusion_matrix(y_true, y_pred).ravel()
    return {
        'True Positive Rate (%)':       round(recall_score(y_true,
y_pred)*100, 2),
        'False Positive Rate (%)':      round(fp/(fp+tn)*100, 2),
```

```
        'Precision (%)':                round(precision_score(y_true, 
y_pred)*100, 2),
        'F-Measure (%)':                round(f1_score(y_true, y_pred)*100, 
2),
        'Relative Absolute Error (%)':  round(abs(y_true-
y_pred).sum()/abs(y_true-y_true.mean()).sum()*100, 2),
        'AUC (%)':                      round(roc_auc_score(y_true, 
y_prob)*100, 2),
        'Kappa Statistics (%)':         round(cohen_kappa_score(y_true, 
y_pred)*100, 2),
        'Accuracy (%)':                 round(accuracy_score(y_true, 
y_pred)*100, 2)
    }

comparison_table = pd.DataFrame({
    'Metric':                    evaluate_model(y_test, ridge_pred, 
ridge_prob).keys(),
    'Ridge Logistic Regression': evaluate_model(y_test, ridge_pred, 
ridge_prob).values(),
    'Random Forest':             evaluate_model(y_test, rf_pred, 
rf_prob).values(),
    'XGBoost':                   evaluate_model(y_test, xgb_pred, 
xgb_prob).values()
})
print(comparison_table)
```

E.9 Multiclass Classification Models (5-Class Vaccination Score)

E.9.1 Multiclass Train–Test Split

```
X = df_model[features]
y = df['vaccinated_5class']
print(y.value_counts())

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42, stratify=y)
```

E.9.2 Multinomial Ridge Logistic Regression (C = 10)

Multinomial logistic regression with L2 penalty. Best C=10 selected via 5-fold GridSearchCV (scoring: weighted F1).

```
final_model = LogisticRegression(
    penalty='l2', C=10, solver='lbfgs', 
multi_class='multinomial', max_iter=5000, random_state=42)
final_model.fit(X_train, y_train)

test_pred = final_model.predict(X_test)
print(classification_report(y_test, test_pred))
```

E.9.3 Multinomial Ridge LR with Class Weights (C = 0.01)

Same model as above but with class_weight="balanced" to handle class imbalance. Best C=0.01 from GridSearchCV. Train-test split: 80/20.

```
# Train-test split (80/20 for this model)
```

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y)

ridge_multi = LogisticRegression(
    penalty='l2', C=0.01, solver='lbfgs',
multi_class='multinomial', class_weight='balanced',
max_iter=5000, random_state=42)
ridge_multi.fit(X_train, y_train)

test_pred = ridge_multi.predict(X_test)
print(classification_report(y_test, test_pred))
```

E.9.4 Multiclass Random Forest Classifier

Random Forest with balanced class weights. Best hyperparameters: n_estimators=200, max_depth=10, min_samples_split=2, max_features=sqrt. Train-test split: 70/30.

```
# Reset to 70/30 split
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42, stratify=y)

rf_model = RandomForestClassifier(
n_estimators=200, max_depth=10,
min_samples_split=2, min_samples_leaf=1,
max_features='sqrt', class_weight='balanced', random_state=42)
rf_model.fit(X_train, y_train)

test_pred = rf_model.predict(X_test)
test_prob = rf_model.predict_proba(X_test)
print(classification_report(y_test, test_pred))

# Feature Importance
importance_df = pd.DataFrame({'Variable': X_train.columns,
                              'Importance':
rf_model.feature_importances_})
print(importance_df.sort_values('Importance', ascending=False))
```

E.9.5 Multiclass XGBoost Classifier

Gradient boosted trees for multiclass output. Best hyperparameters: n_estimators=200, max_depth=5, learning_rate=0.1, subsample=0.8, gamma=0.1.

```
xgb_model = XGBClassifier(
    objective='multi:softprob',
num_class=len(np.unique(y)),
eval_metric='mlogloss',
n_estimators=200, max_depth=5, learning_rate=0.1,
    subsample=0.8, colsample_bytree=1.0, gamma=0.1,
random_state=42)
xgb_model.fit(X_train, y_train)

test_pred = xgb_model.predict(X_test)
test_prob = xgb_model.predict_proba(X_test)
print(classification_report(y_test, test_pred))
```

E.10 Multiclass ROC-AUC Comparison

```
from sklearn.preprocessing import label_binarize
from sklearn.metrics import roc_auc_score, roc_curve, auc

classes    = np.unique(y_test)
y_test_bin = label_binarize(y_test, classes=classes)

ridge_prob = ridge_multi.predict_proba(X_test)
rf_prob    = rf_model.predict_proba(X_test)
xgb_prob   = xgb_model.predict_proba(X_test)

ridge_macro_auc = roc_auc_score(y_test_bin, ridge_prob, multi_class='ovr',
average='macro')
rf_macro_auc    = roc_auc_score(y_test_bin, rf_prob,   multi_class='ovr',
average='macro')
xgb_macro_auc   = roc_auc_score(y_test_bin, xgb_prob,  multi_class='ovr',
average='macro')

def compute_micro_roc(y_true, y_prob):
fpr, tpr, _ = roc_curve(y_true.ravel(), y_prob.ravel())
    return fpr, tpr, auc(fpr, tpr)

ridge_fpr, ridge_tpr, ridge_micro_auc = compute_micro_roc(y_test_bin,
ridge_prob)
rf_fpr,   rf_tpr,   rf_micro_auc    = compute_micro_roc(y_test_bin,
rf_prob)
xgb_fpr,   xgb_tpr,   xgb_micro_auc  =compute_micro_roc(y_test_bin,
xgb_prob)

roc_summary = pd.DataFrame({
    'Model':         ['Ridge Multinomial', 'Random Forest', 'XGBoost'],
    'Macro ROC-AUC': [round(ridge_macro_auc,4), round(rf_macro_auc,4),
round(xgb_macro_auc,4)],
    'Micro ROC-AUC': [round(ridge_micro_auc,4), round(rf_micro_auc,4),
round(xgb_micro_auc,4)]
})
print(roc_summary)
```